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Department of Electrical and System Engineering
Digital Integrated Circuits AND VLSI Fundamentals

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Baseline Design

GATE Schematics

AND GATE SCHEMATIC

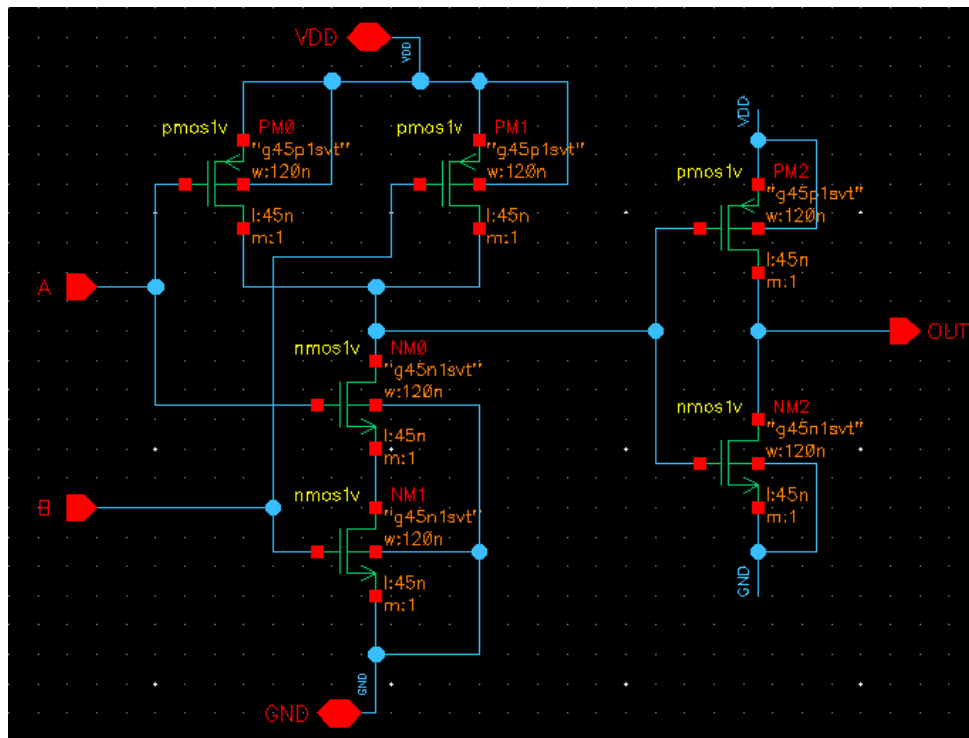


Figure 1 - AND GATE Schematic

AND SYMBOL

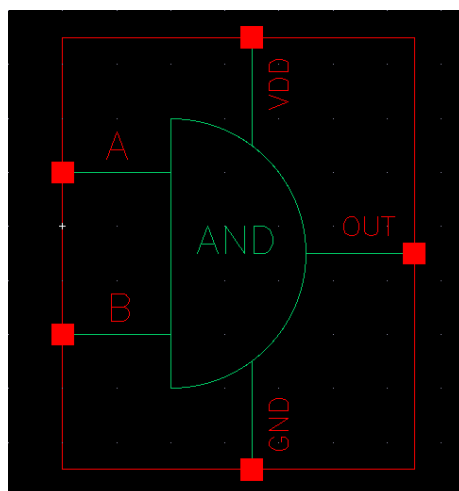


Figure 2 - AND GATE Symbol

OR GATE SCHEMATIC

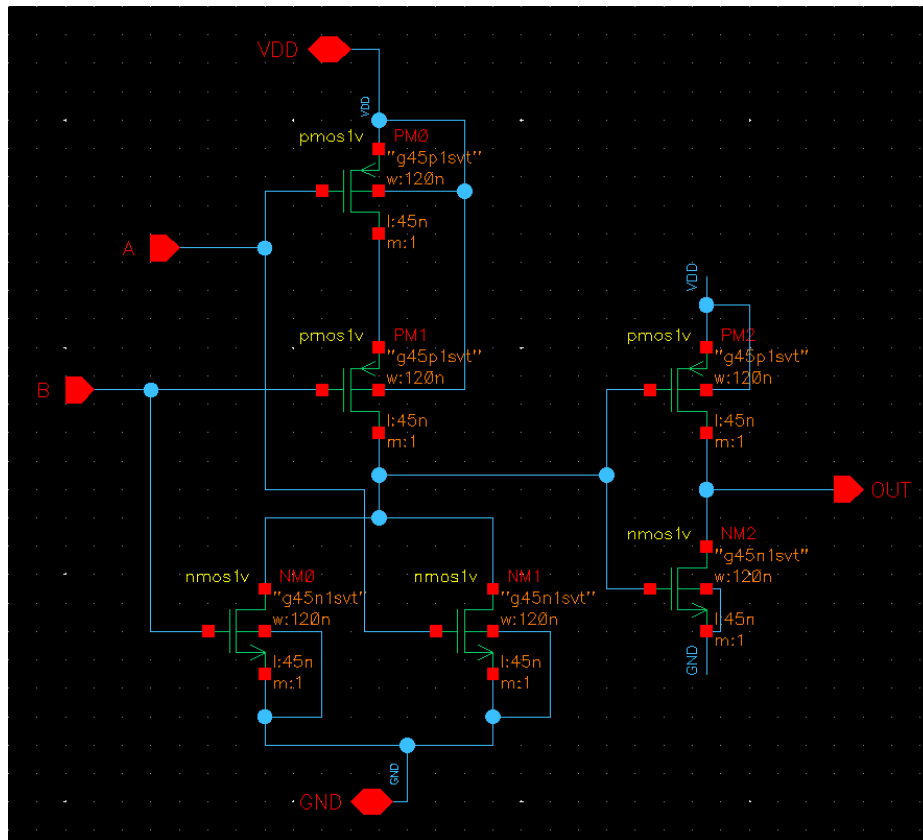


Figure 3 - OR GATE SCHEMATIC

OR SYMBOL

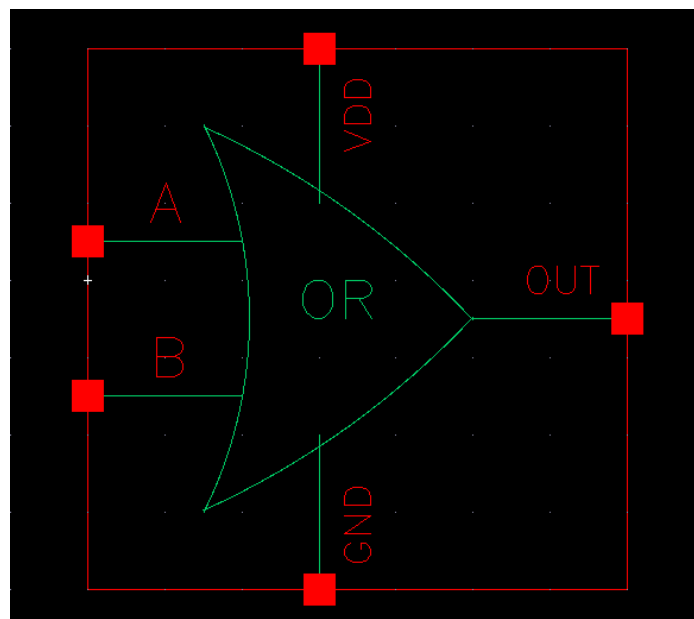


Figure 4 - OR GATE Symbol

XOR GATE SCHEMATIC

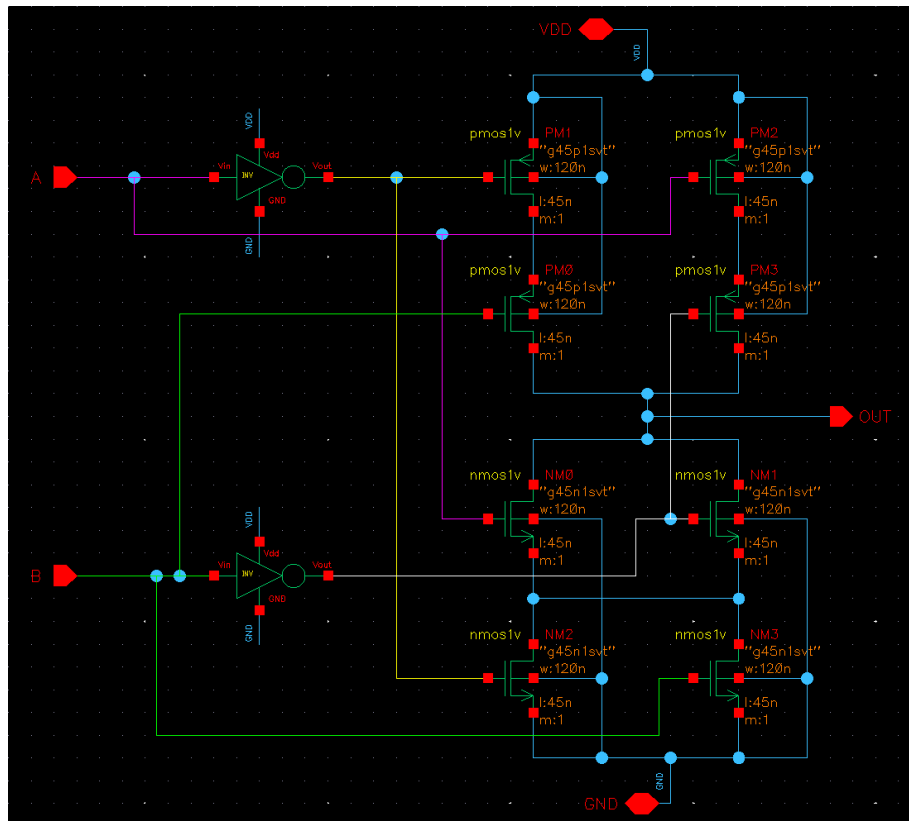


Figure 5 - XOR GATE Schematic

XOR SYMBOL

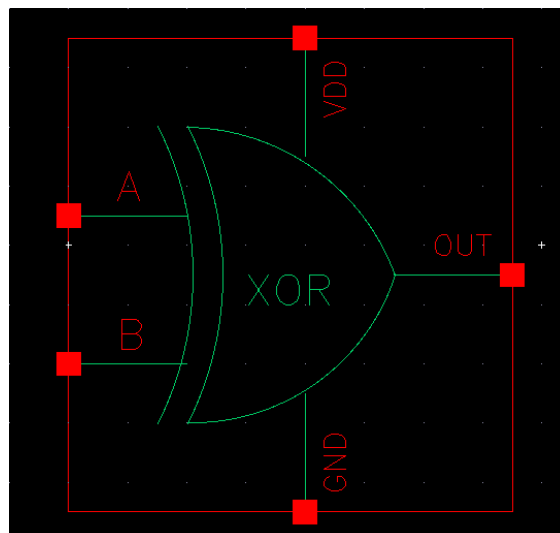


Figure 6 - XOR GATE Symbol

Bit-Slice Cell Design for Baseline Design

Each full adder bit-slice in the RCA is implemented using **48 CMOS transistors**, with the gates sized to minimum dimensions for the **45nm process**. Specifically, the inverter in each bit-slice is sized with a **width (W) of 120nm and length (L) of 45nm** for both the PMOS and NMOS transistors, in line with the minimum feature sizes for this process node.

The full adder uses basic CMOS logic:

- **XOR gates** are employed to generate the sum output.
- **AND and OR gates** are used to compute the carry-out, which is then passed to the next full adder in the chain.

This configuration ensures that the adder is compact, utilizing the smallest possible transistor sizes while still providing reliable logic functionality.

Design Challenges and Considerations

The primary challenge in the Ripple-Carry Adder design is its **linear delay**. Since the carry must ripple through each of the **8 stages**, the delay grows linearly with the number of bits, making it slower for larger bit-width adders. The propagation delay of the carry signal across all stages limits the overall speed of the adder. This delay is particularly significant when the carry chain is long, as the worst-case delay occurs when the carry signal ripples from the least significant to the most significant bit.

Additionally, transistor sizing is a critical factor for ensuring the adder operates efficiently. The **minimum-sized transistors (W=120nm, L=45nm)** used in the design are optimized for area but must still drive the capacitance of the full adder bit-slices, especially in a cascaded structure. Proper sizing of the pull-up (PMOS) and pull-down (NMOS) networks is crucial for balancing delay and power consumption.

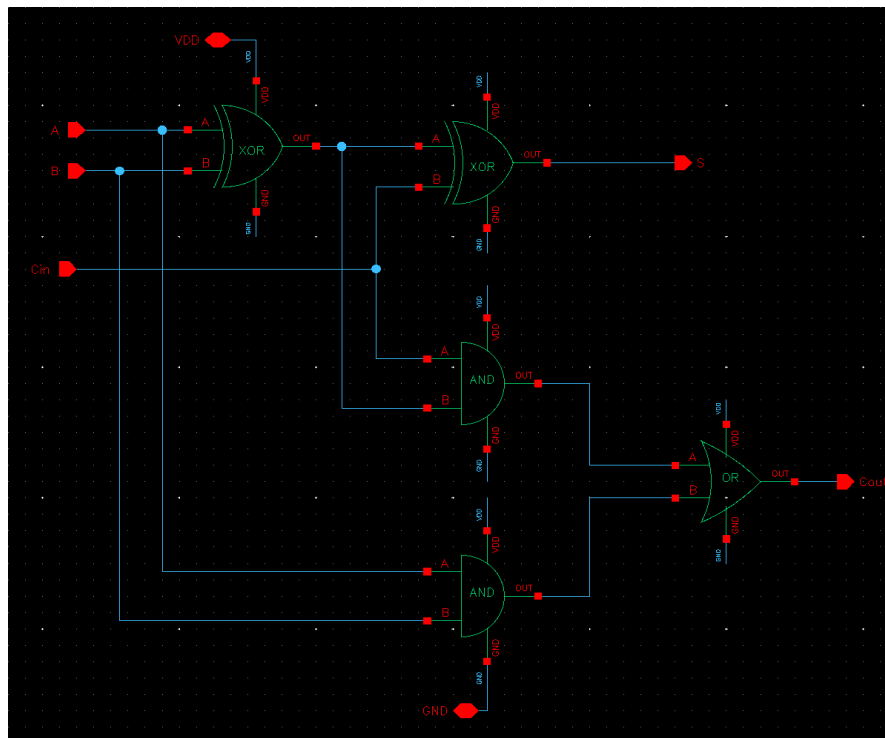


Figure 7 - Full Adder Schematic

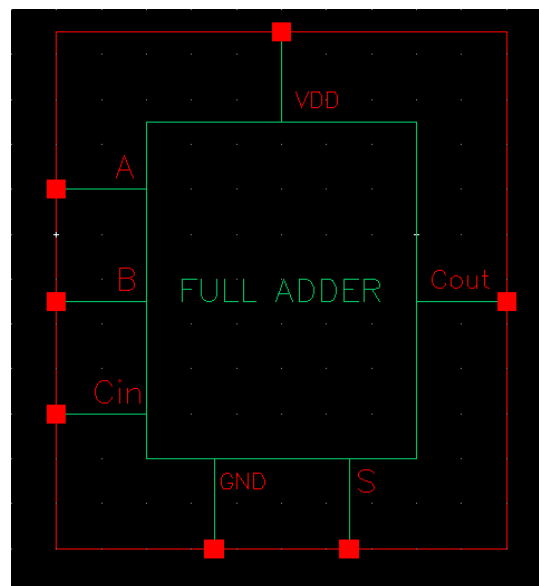


Figure 8 - Full Adder Symbol

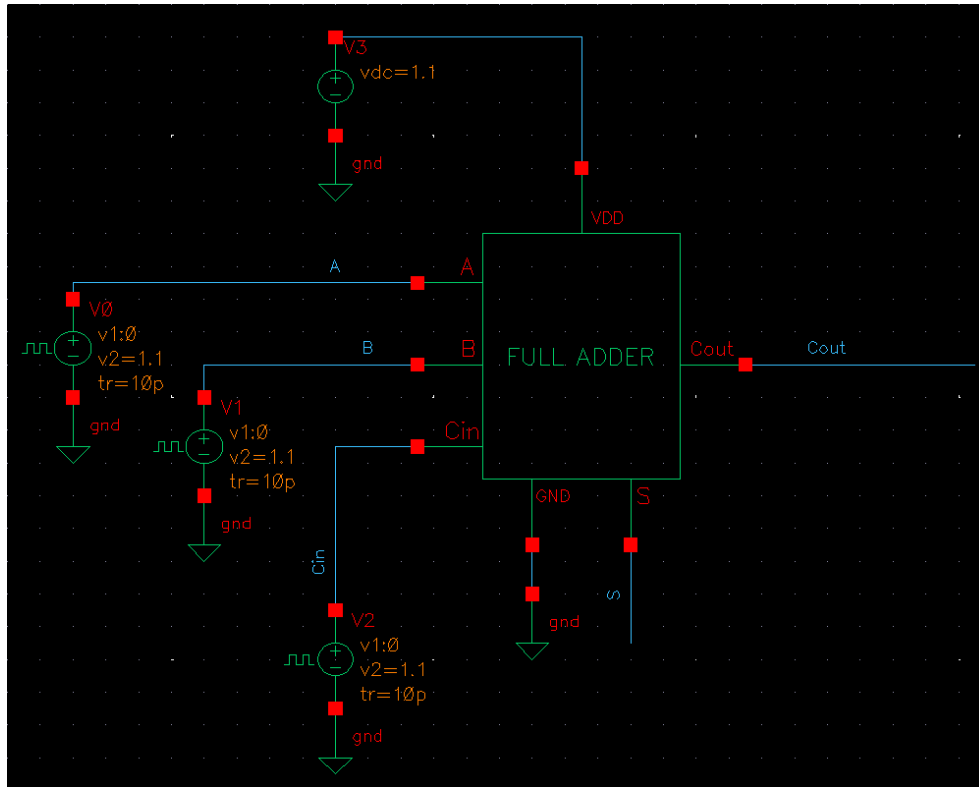


Figure 9 - Full Adder Test Schematic

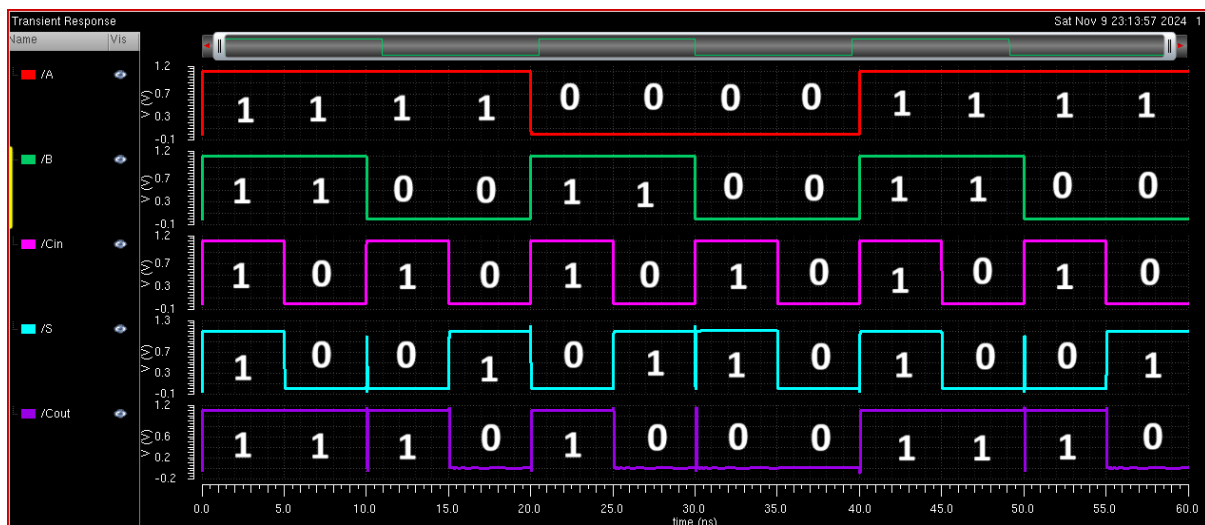


Figure 10 - Full Adder Test Graph

The waveform shows the transient response of a 1-bit full adder circuit, where A and B are the primary inputs, Cin is the carry input, and S and Cout represent the sum and carry-out outputs, respectively.

To test the functionality of the 1-bit adder slice, different periods for each input are chosen such that all 8 possible input combinations are covered. The input pulse width is set with a 1:2 ratio (50% duty cycle) relative to the period.

The sum S (cyan) follows the equation $S = A \oplus B \oplus \text{Cin}$, while the carry-out Cout (magenta) is generated according to $\text{Cout} = (A \cdot B) + (\text{Cin} \cdot (A \oplus B))$.

The outputs Sum and Cout align with expected values, confirming correct operation across all 8 test cases.

Baseline Design Schematic

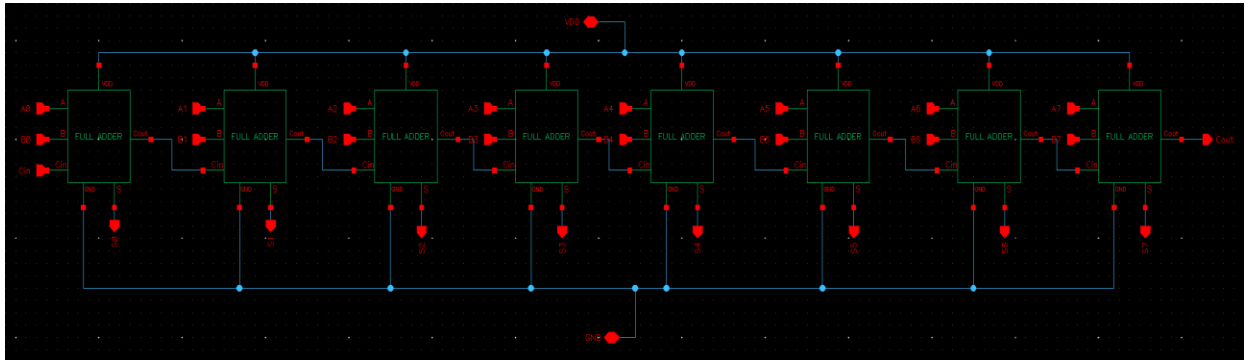


Figure 11 - Baseline Design Schematic

Baseline Design Description

The baseline design for the 8-bit adder follows the **Ripple-Carry Adder (RCA)** topology, which consists of **8 full adder** cells arranged in a cascaded structure. Each full adder performs a **1-bit addition with a carry-in and produces a sum and carry-out**, which is passed to the next bit-slice.

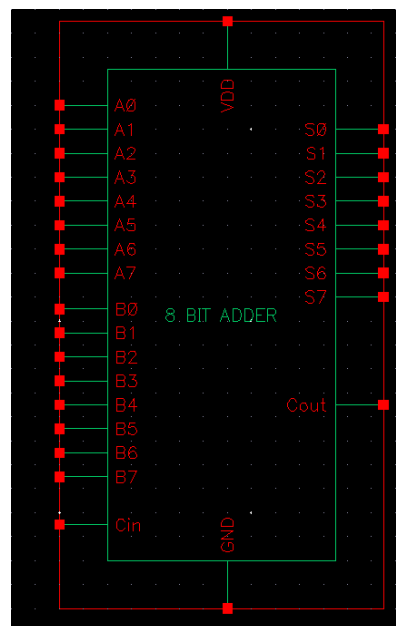


Figure 12 - Baseline Symbol

Baseline Design Test Schematic

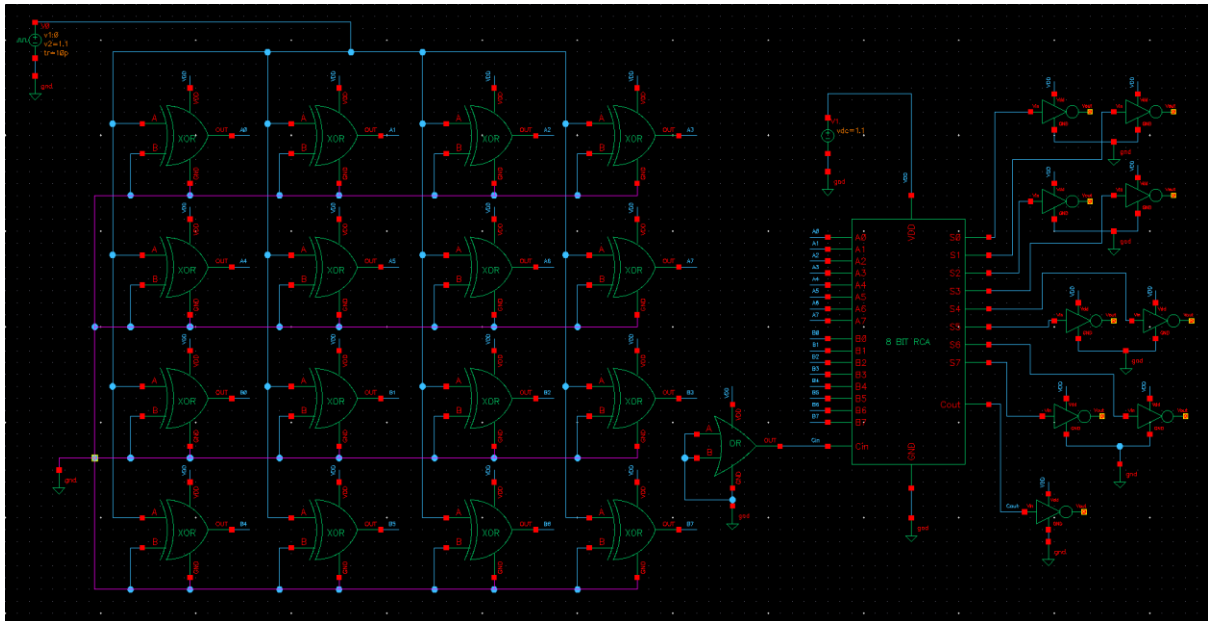


Figure 13 - Baseline Design Test Schematic

Verification of 8-bit Baseline Adder for Logical Correction

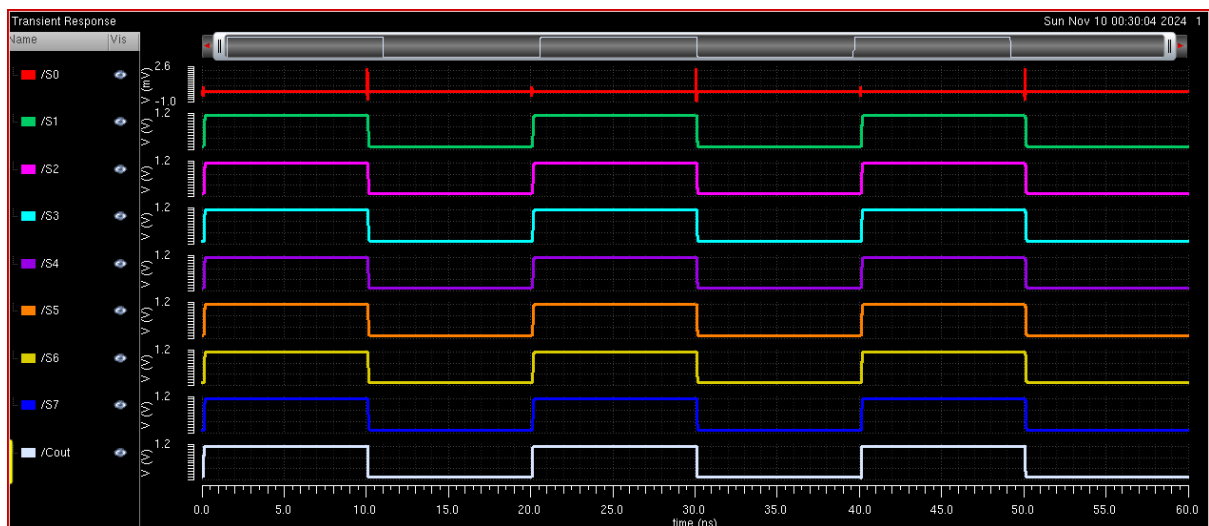


Figure 14 - Test Graph for Verification of 8-bit Baseline Adder

To verify the logical correctness of the 8-bit baseline adder, we tested it by switching both the inputs from all 0s (**00000000**) to all 1s (**11111111**). This represents a full-range input transition, allowing to observe the behavior of carry propagation across all bits.

The waveform in the transient response confirms that the sum outputs (S0 to S7) and the carry-out (Cout) behave as expected, indicating that the adder correctly handles this edge case and ensures accurate addition across its entire input range.

Delay Test Case Description and Measurement

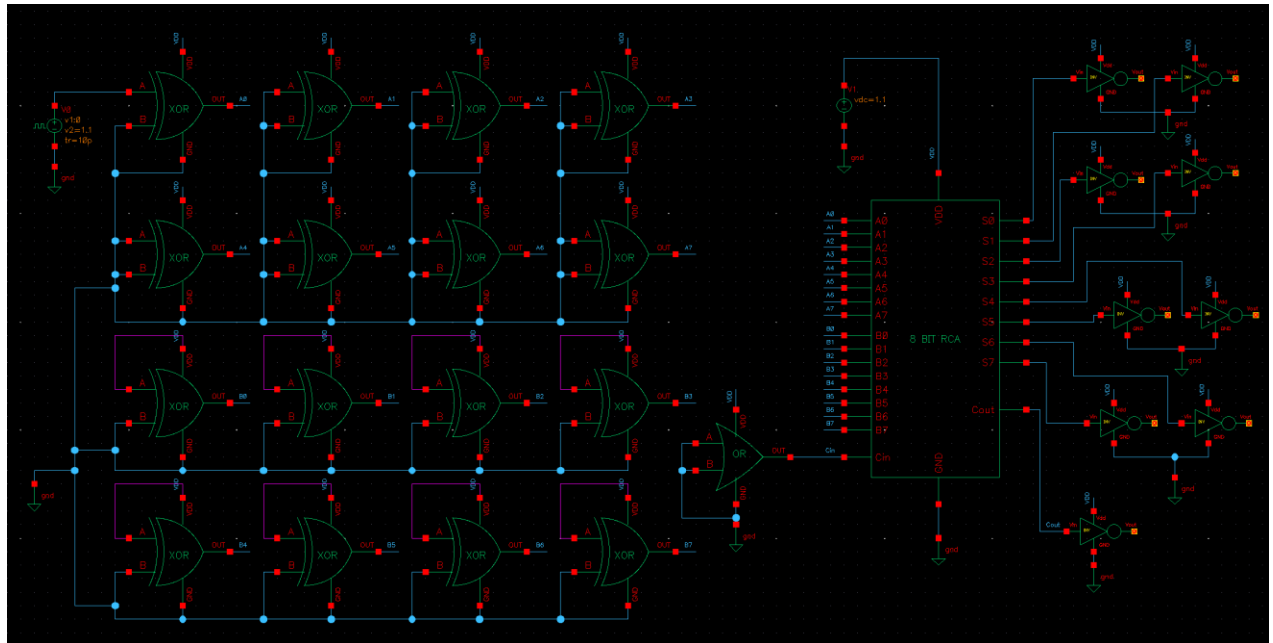


Figure 15 - Delay Test Schematic

For the worst-case scenario, we have selected the following as the input case:

A0: 0 → 1 (switches from 0 to 1)

A1- A7: 0

B0 - B7: 1 (VDD)

Cin: 0

The following add function is implemented:

A:	0 0 0 0 0 0 0 (0→1)
B:	1 1 1 1 1 1 1
Cin:	0
	1 0 0 0 0 0 0 0

This input case ensures that the carry propagates through every sum output stage. This propagation adds delay to the output every step the output is generated.

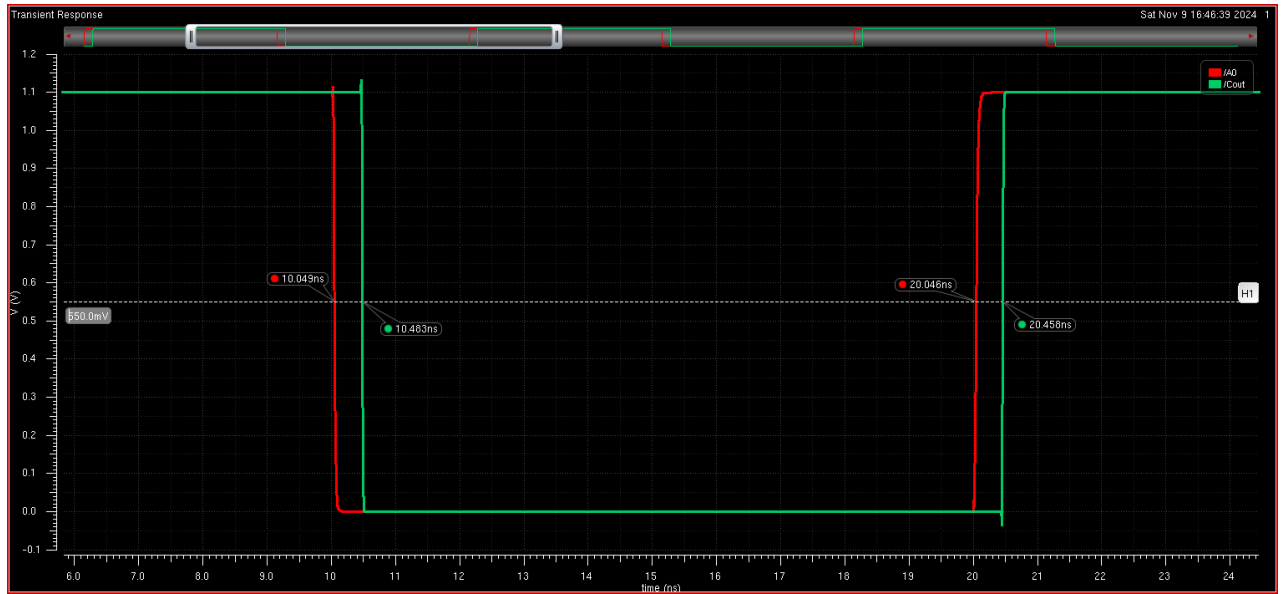


Figure 16 - Baseline Design Delay Test Graph

Charging Delay, $\tau_{pLH} = (20.458 - 20.046) \text{ ns} = \mathbf{0.412 \text{ ns}}$

Discharging Delay, $\tau_{pHL} = (10.483 - 10.049) \text{ ns} = \mathbf{0.434 \text{ ns}}$

Avg. Propagation Delay, $\tau_p = \frac{\tau_{pLH} + \tau_{pHL}}{2} = \frac{0.412 + 0.434}{2} = \mathbf{0.423 \text{ ns}}$

So, the obtained worst-case delay is **0.423 ns**.

Baseline Active Energy Max/Avg Measurements

1. Maximum Switching Energy Case

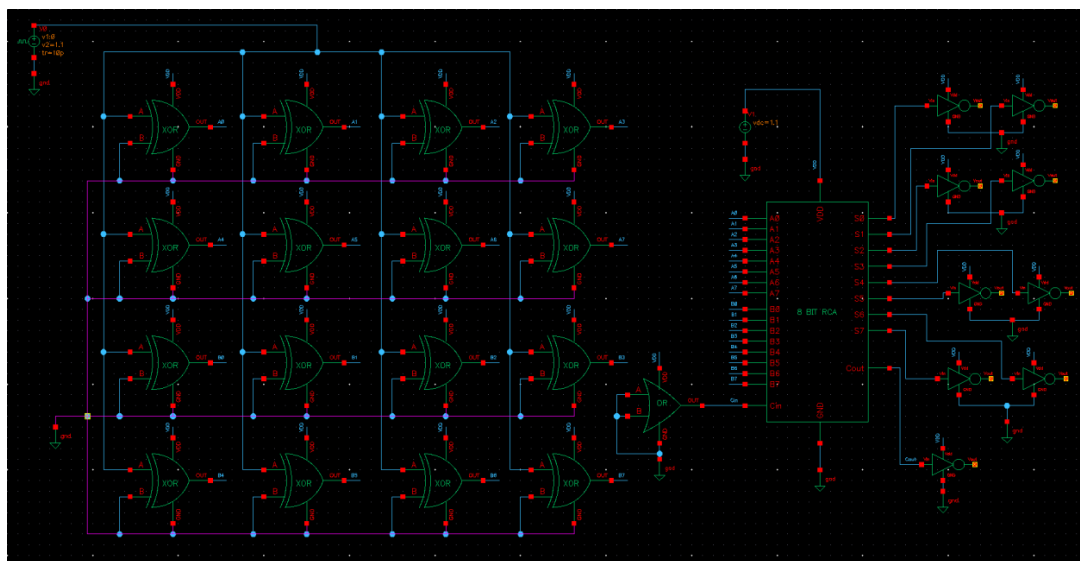


Figure 17 - Baseline Design Max Active Energy Schematic

To maximize switching energy, we simultaneously toggle all A and B inputs from `0` to `1` across both adders. This creates a scenario where every bit—ranging from A0 to A7 and B0 to B7—undergoes a full transition from low to high. By inducing this synchronized state change, we maximize power consumption, capturing the peak switching activity of the system.

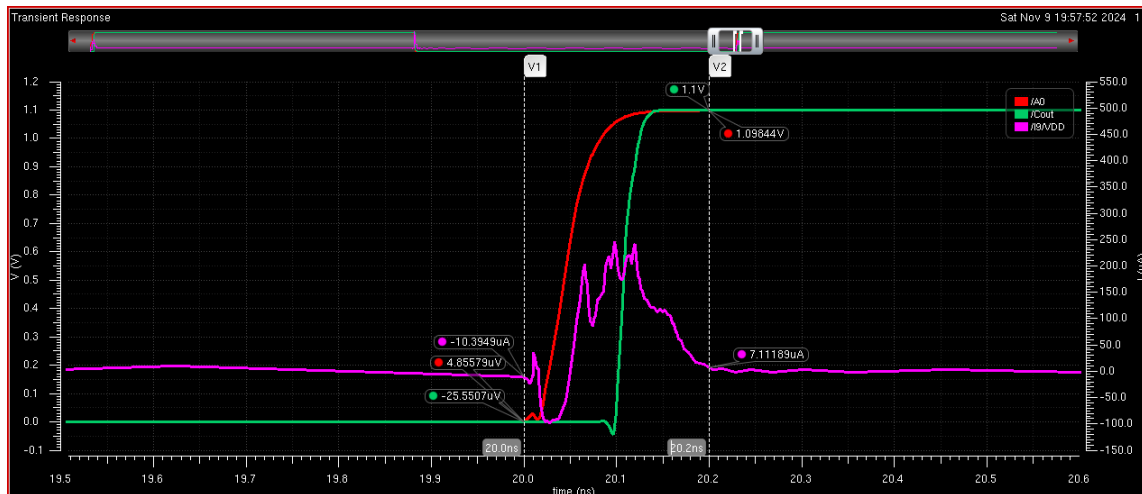


Figure 18 - Baseline Design Max Active Energy Graph

The energy is measured for the waveform where input goes from 0 → 1 as depicted in Figure 18.

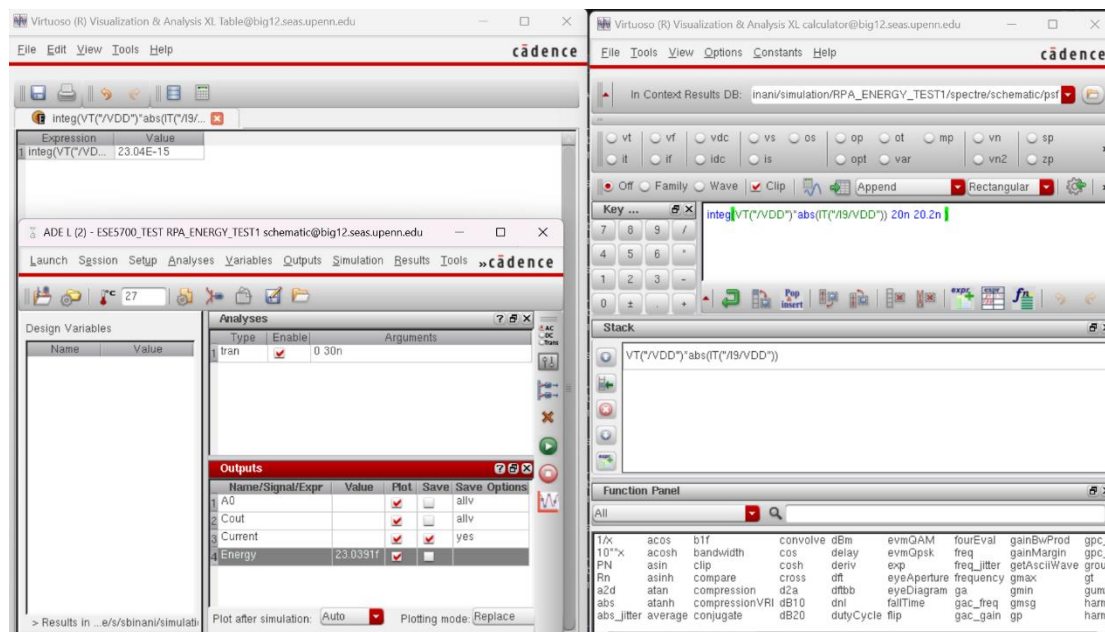


Figure 19 - Baseline Design Max Active Energy

Therefore, the Maximum Switching Energy Value obtained is **23.04 F J**.

2. Average Switching Energy

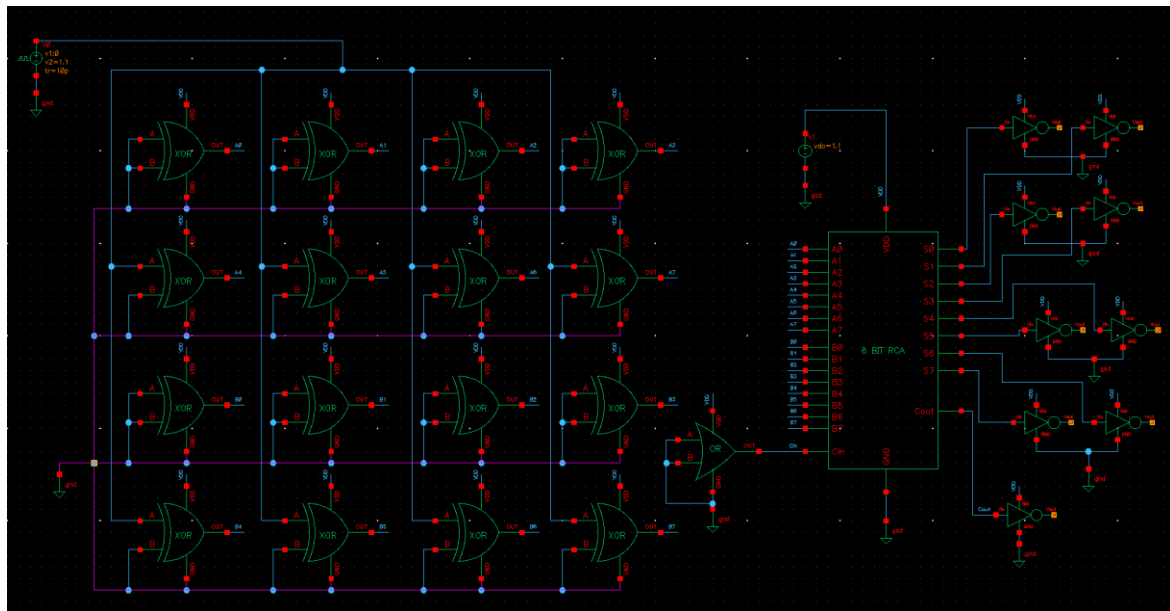


Figure 20 - Baseline Design Average Active Energy Schematic

To calculate the average switching energy, we have configured both A and B inputs as **11110000** for the adders in the first stage. This setup sets the most significant bits (**A4 to A7 and B4 to B7**) to **1** and the lower bits (**A0 to A3 and B0 to B3**) to **0**.

This balanced input pattern ensures partial switching, representing an average case scenario where only half of the bits transition, allowing us to effectively estimate the typical power consumption.

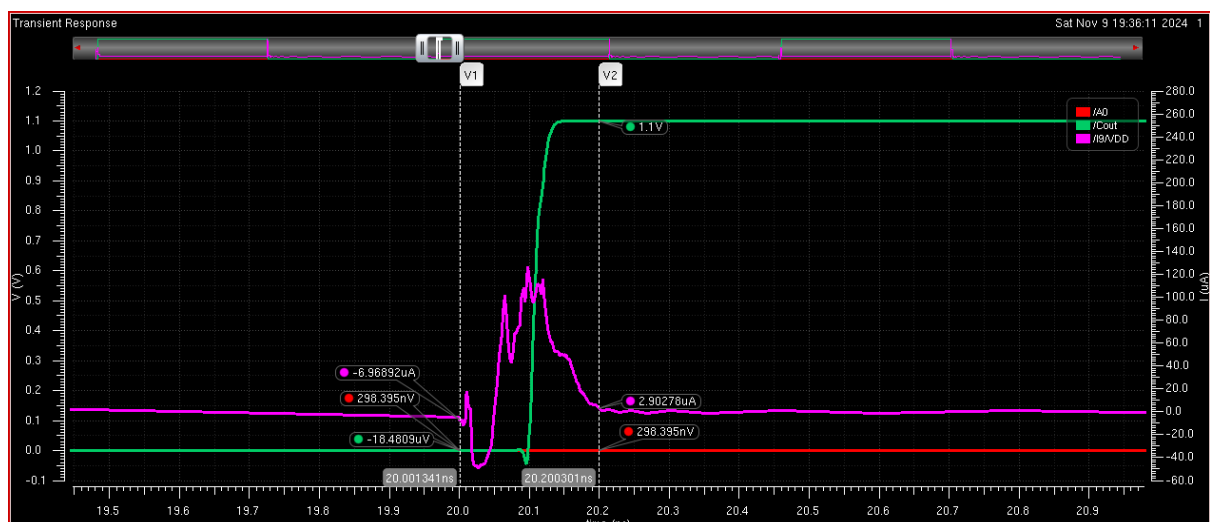


Figure 21 - Baseline Design Average Active Energy Graph

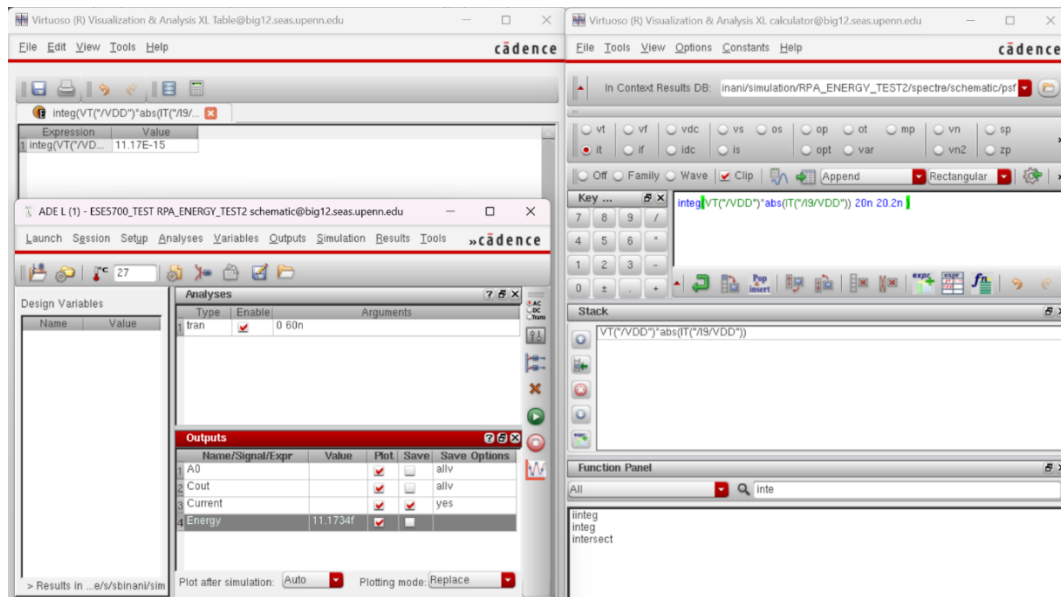


Figure 22 - Baseline Design Average Active Energy

Therefore, the Average Switching Energy Value obtained is **11.17 F J**.

Baseline Leakage Energy Max/Min Cases and Measurements

1. Max Leakage Energy

Max Case – A = 0, B = 0, Cin = 0

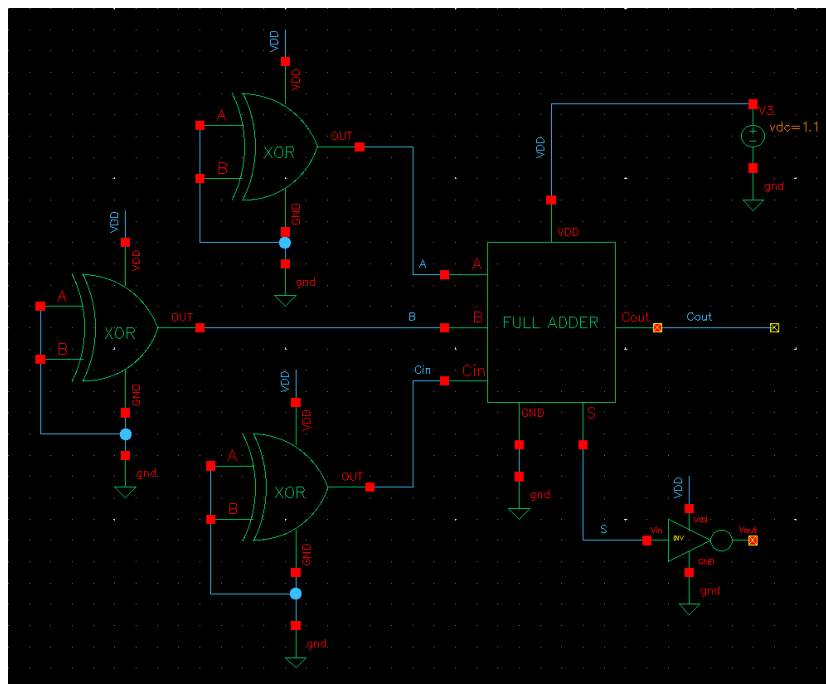


Figure 23 - Baseline Max Leakage Energy Schematic

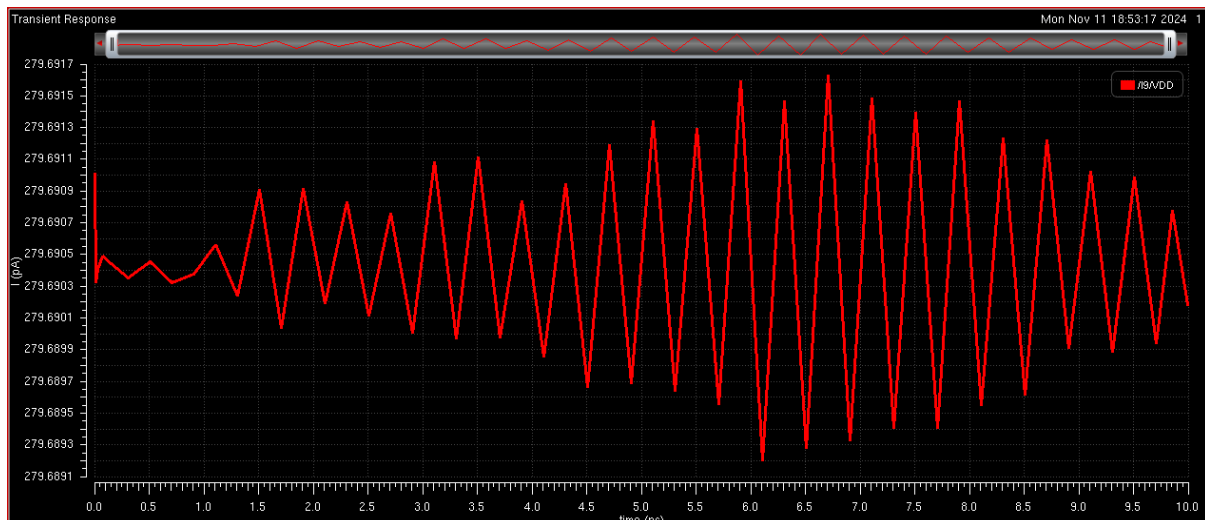


Figure 24 - Baseline Graph for Max Leakage Energy

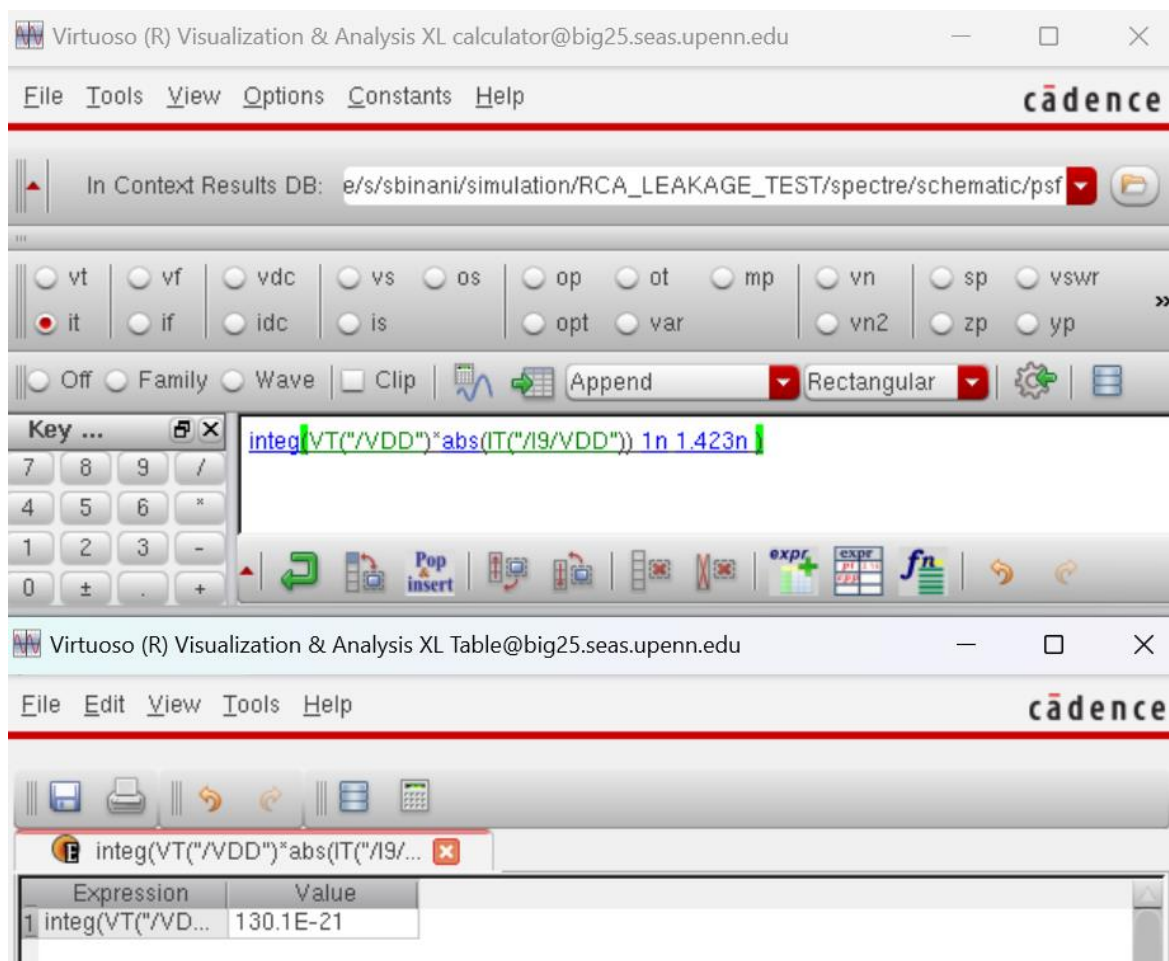


Figure 25 - Max Leakage Energy Calculator

Therefore, the Maximum Leakage Energy Value obtained for bit-slice is 130.10×10^{-21} J.

So, total max leakage energy for 8-bit adder = $130.1 \times 8 = 1040.8 \times 10^{-21}$ J.

2. Minimum Leakage Energy

Min Case – A = 1, B = 1, Cin = 1

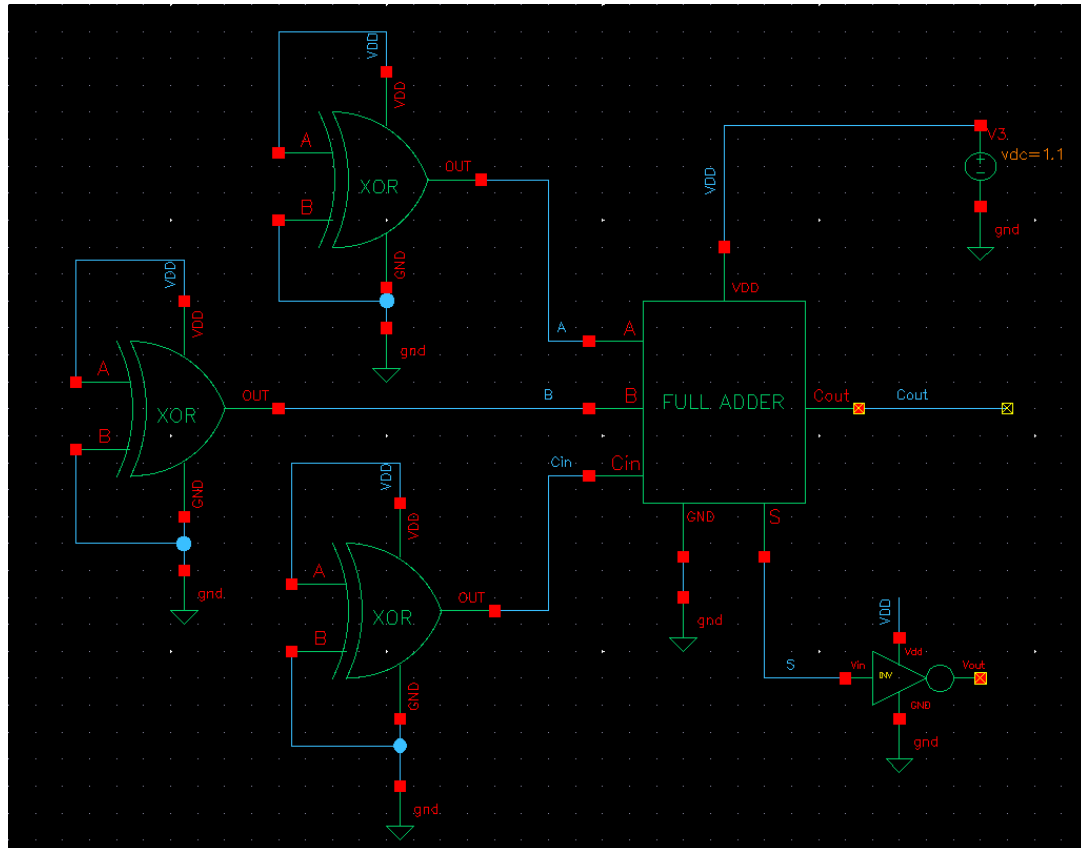


Figure 26 - Min Leakage Energy Schematic

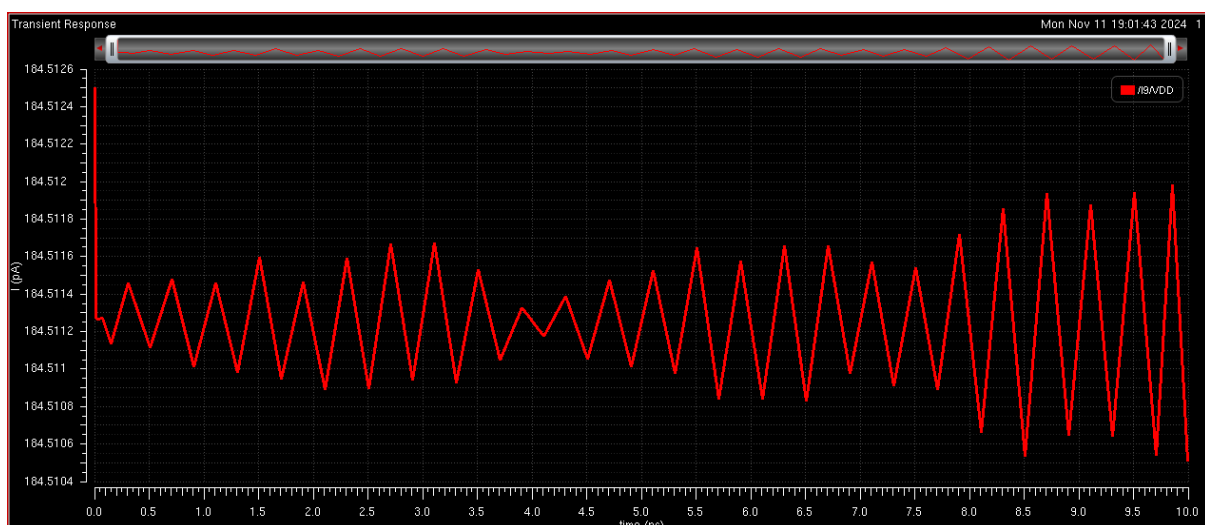


Figure 27 - Min Leakage Energy Graph

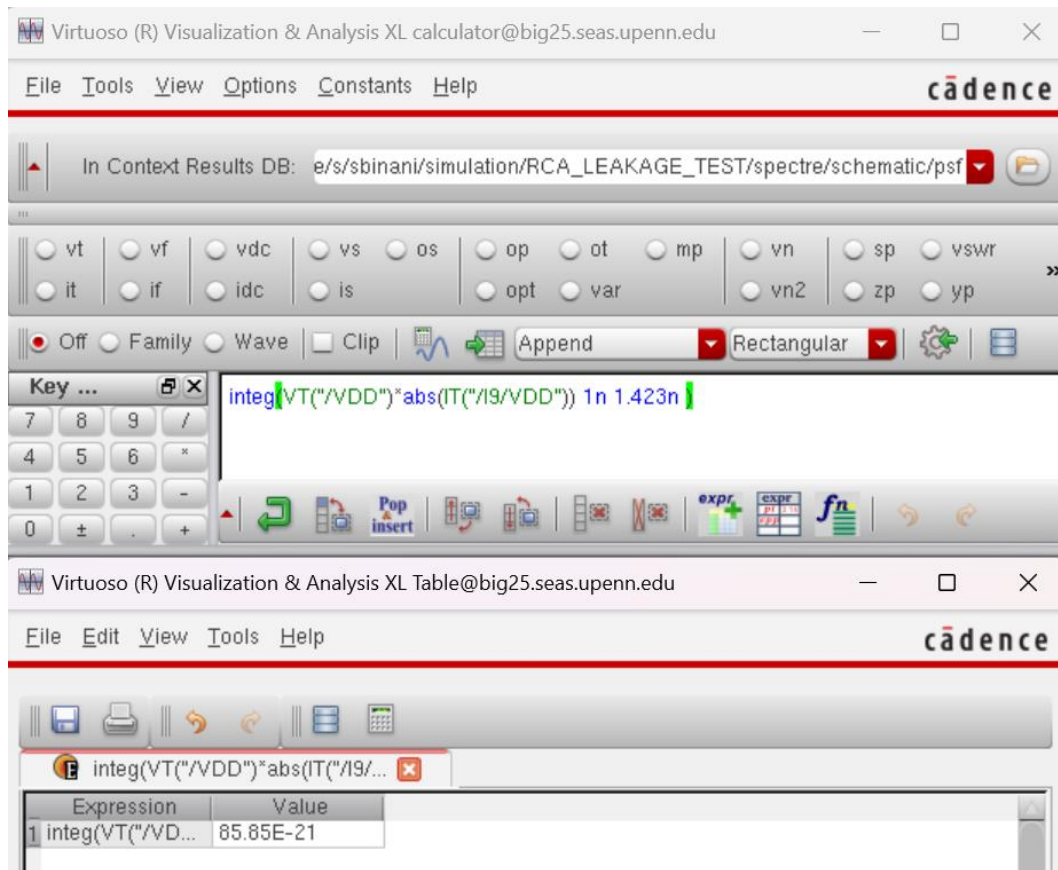


Figure 28 - Min Leakage Energy Calculator

Therefore, the Minimum Leakage Energy Value obtained for bit-slice is **$85.85 \times 10^{-21} \text{ J}$** .

So, total min leakage energy for 8-bit adder = $85.85 \times 8 = \mathbf{686.8 \times 10^{-21} \text{ J}}$.

Optimized Design

2:1 MUX SCHEMATIC

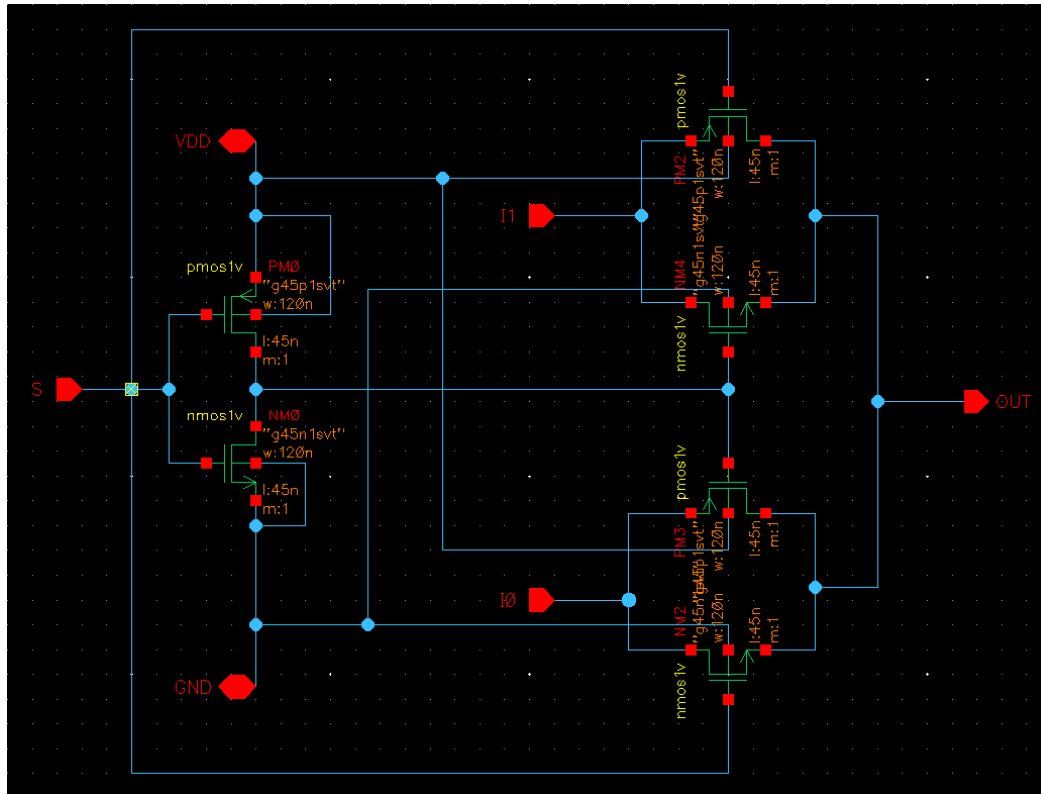


Figure 29 - 2:1 MUX Schematic

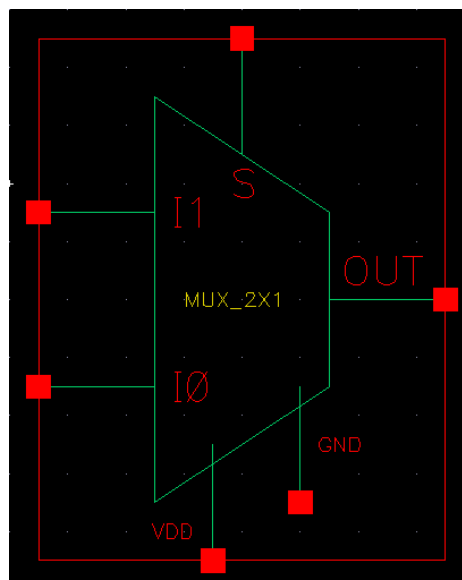


Figure 30 - 2:1 MUX Symbol

NOR GATE

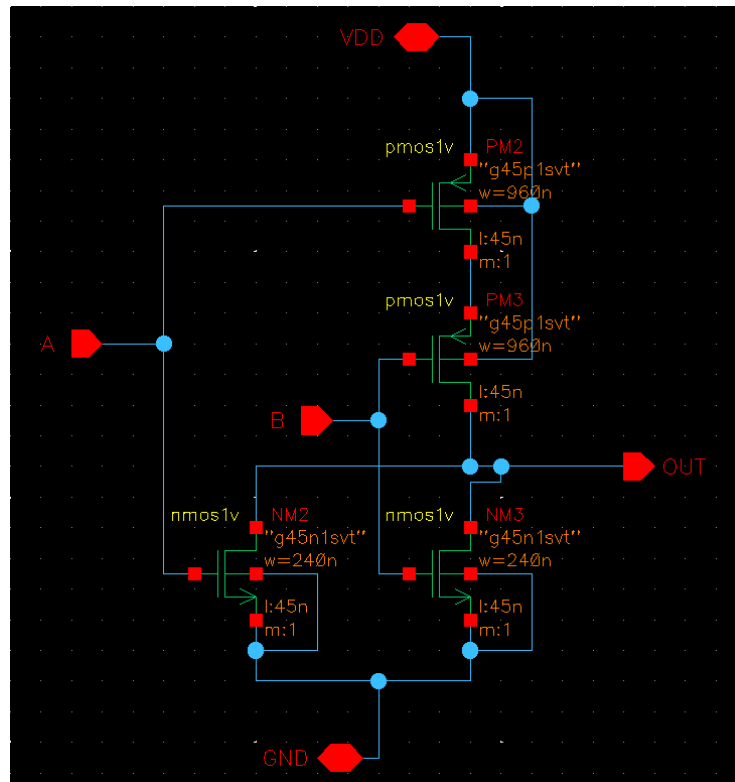


Figure 31 - NOR Gate Schematic

NOR GATE SYMBOL

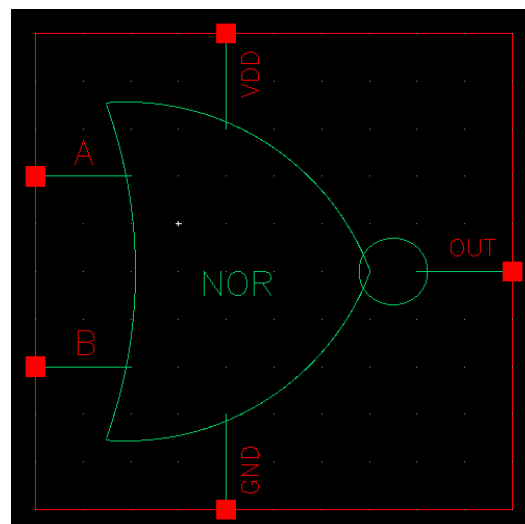


Figure 32 - NOR Gate Symbol

AND4 GATE

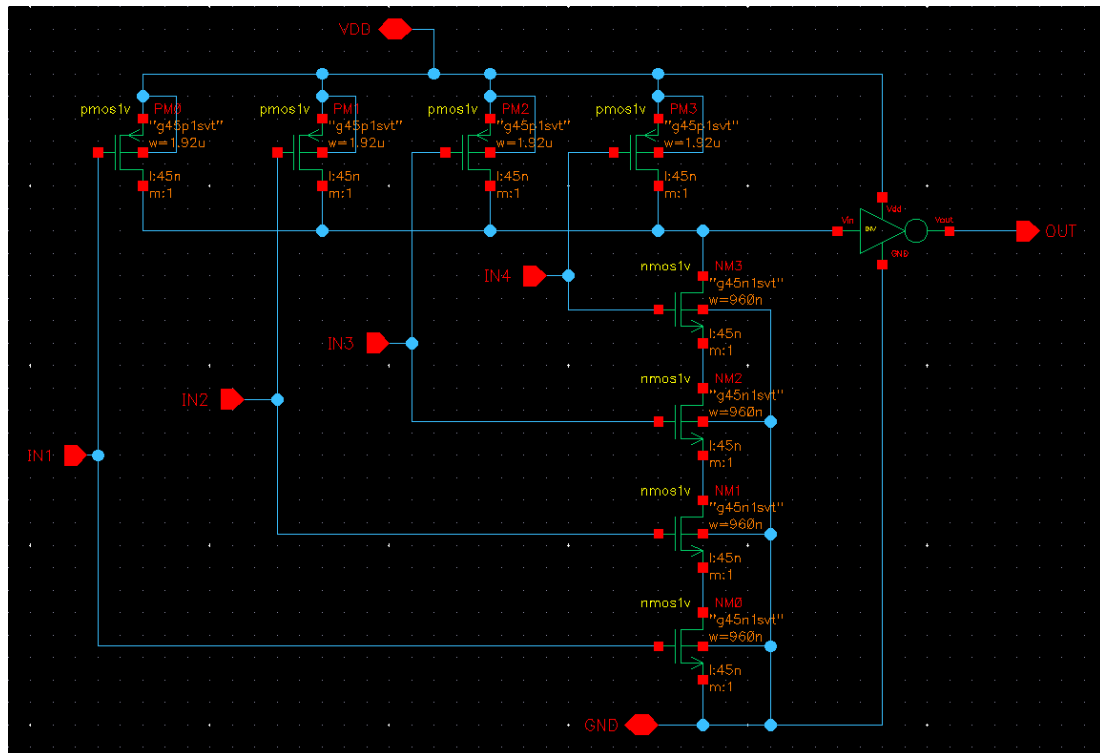


Figure 33 – AND4 Gate Schematic

AND4 GATE SYMBOL

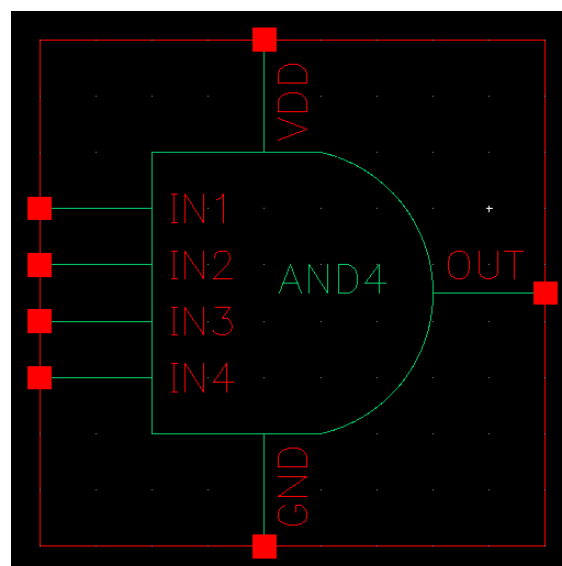


Figure 34 – AND4 Gate Symbol

Bit-Slice Cell Design for Optimized Design

Each full adder bit-slice in the RCA is implemented using **42 CMOS transistors**, with the gates sized to minimum dimensions for the **45nm process**. Specifically, the inverter in each bit-slice is sized with a **width (W) of 120nm** and **length (L) of 45nm** for both the PMOS and NMOS transistors, in line with the minimum feature sizes for this process node.

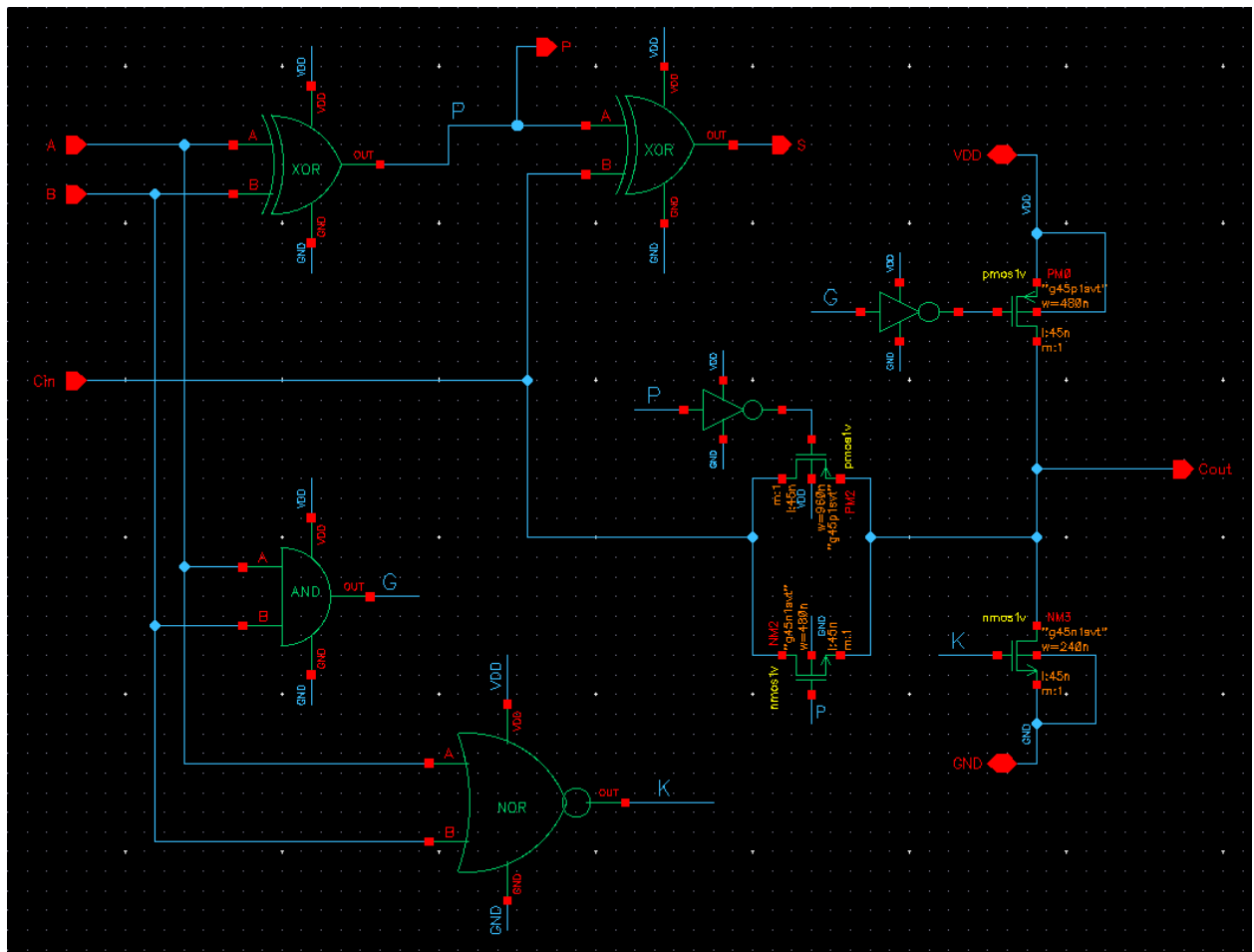


Figure 35 - Full Adder Schematic for Optimized Design

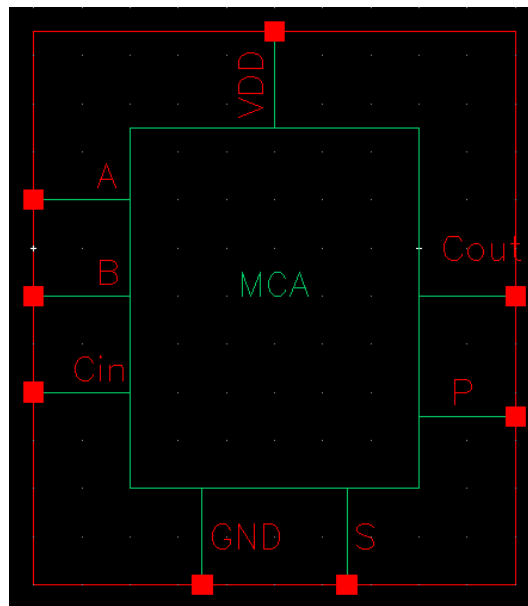


Figure 36 - Full Adder Symbol for Optimized Design

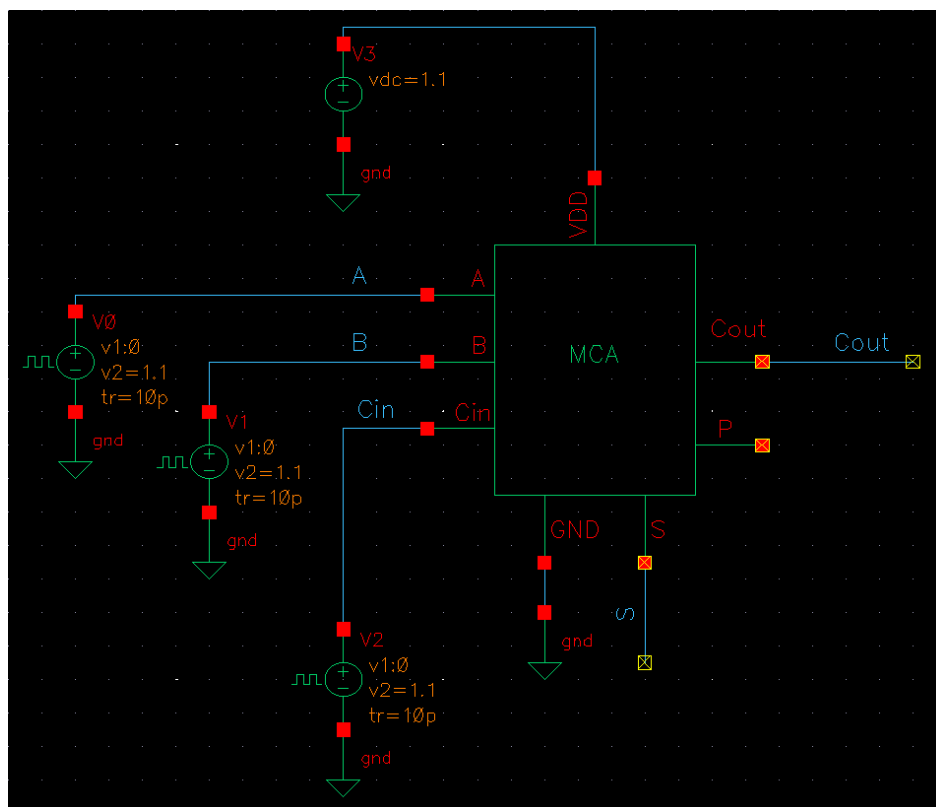


Figure 37 - Full Adder Test Schematic for Optimized Design

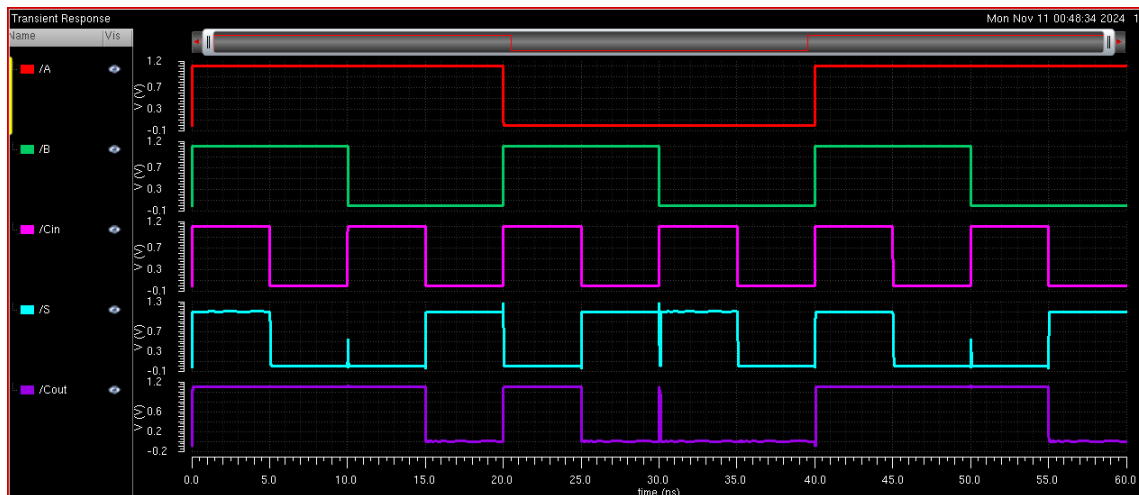


Figure 38 - Graph for bit-slice Full Adder

Optimized Design Schematic

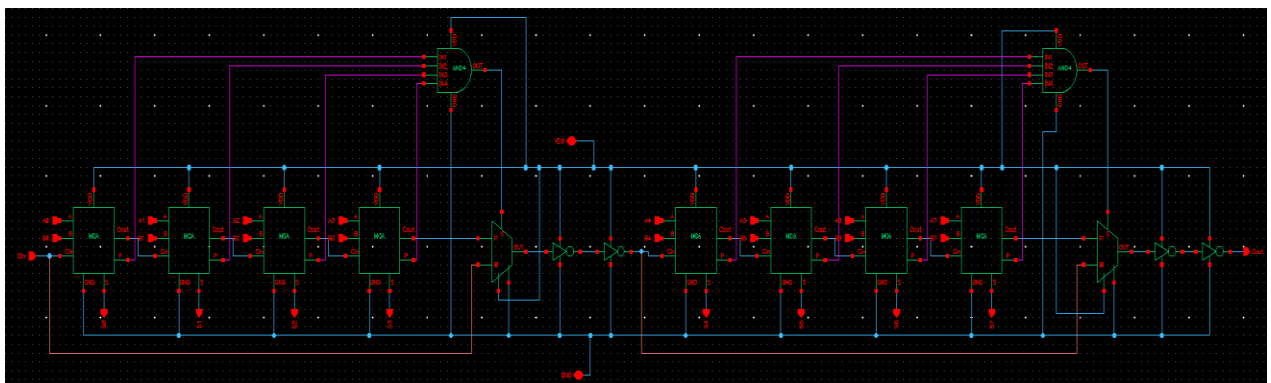


Figure 39 - Optimized Design Schematic

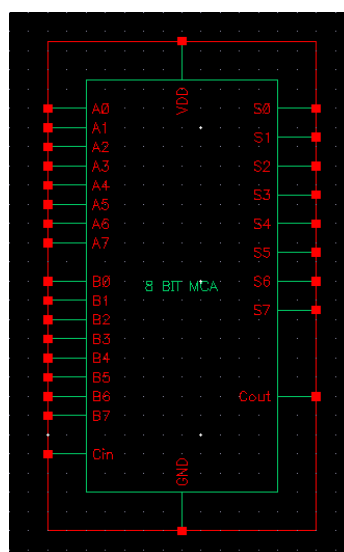


Figure 40 - Optimized Design Symbol

Optimized Design Description

Our optimized 8-bit adder leverages the **Manchester Carry Adder (MCA) architecture with Carry Bypass**, where we have implemented 8 modified full adder cells in a hierarchical carry chain structure for faster carry propagation. The carry propagates either through the bypass path, or a carry is generated through the chain. We chose this design over the traditional RCA because it **processes carry signals in parallel through our Manchester carry chain**, which will drastically reduce our critical path delay.

We used **2:1 multiplexer in the carry chain** which proved crucial for optimizing performance. These multiplexers enabled faster carry select logic, allowing us to compute carry values in advance and select the correct path dynamically. By placing these multiplexers between our modified full adder stages, we achieved efficient carry propagation through parallel path selection, fundamentally reducing the critical delay.

Table 1 - Carry Status Table

A	B	Cin	S	Cout	Carry Status
0	0	0	0	0	Kill
0	0	1	1	0	Kill
0	1	0	1	0	Propagate
0	1	1	0	1	Propagate
1	0	0	1	0	Propagate
1	0	1	0	1	Propagate
1	1	0	0	1	Generate
1	1	1	1	1	Generate

$$G = A.B$$

$$P = A \oplus B$$

$$K = NOT(A + B)$$

$$S = A \oplus B \oplus Cin = P \oplus Cin$$

$$Cout = G + P.Cin$$

Optimized Design Test Schematic

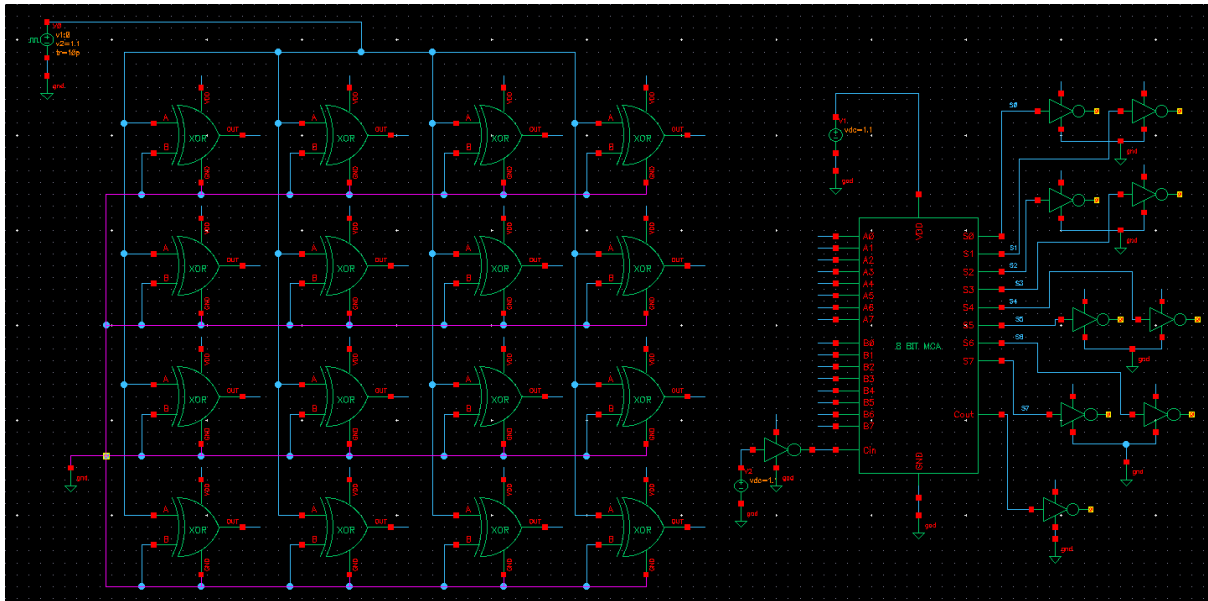


Figure 41 - Optimized Design Test Schematic

Verification of 8-bit Optimized Adder for Logical Correction

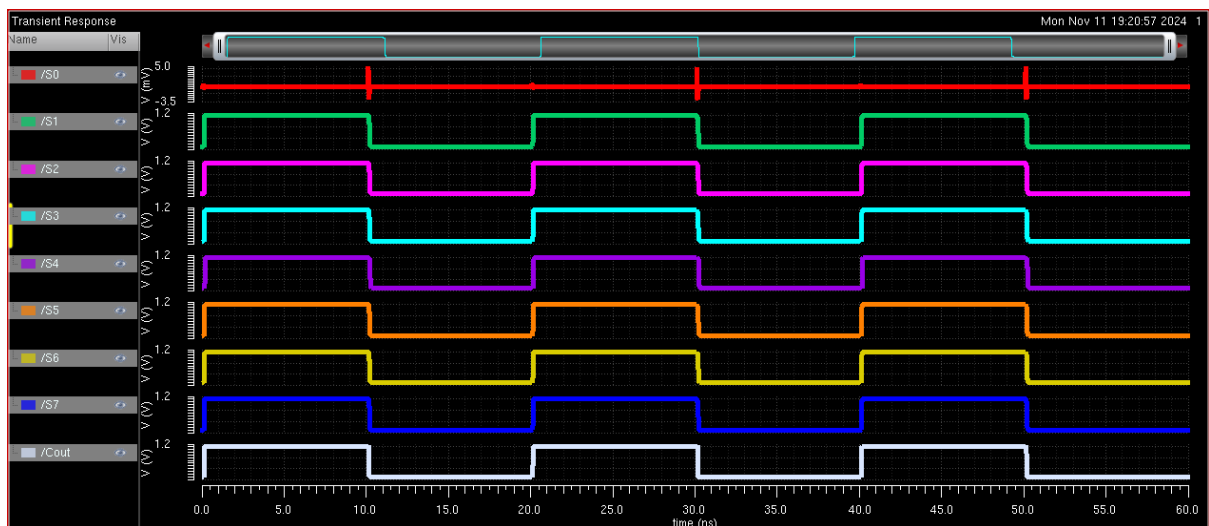


Figure 42 - Test Graph for Verification of 8-bit Optimized Adder

To verify the logical correctness of the 8-bit optimized adder, we tested it by switching the inputs from all 0s (**00000000**) to all 1s (**11111111**). This represents a full-range input transition, allowing to observe the behavior of carry propagation across all bits.

The waveform in the transient response confirms that the sum outputs (S0 to S7) and the carry-out (Cout) behave as expected, indicating that the adder correctly handles this edge case and ensures accurate addition across its entire input range.

So, the obtained worst-case delay is **0.29575 ns**.

Thus, we see from the calculations that the sizing done for optimized design has resulted in reducing the delay from 0.423 ns to 0.29575 ns which is about 30 % reduction in delay.

Optimized Active Energy Max/Avg Measurements

1. Maximum Switching Energy Case

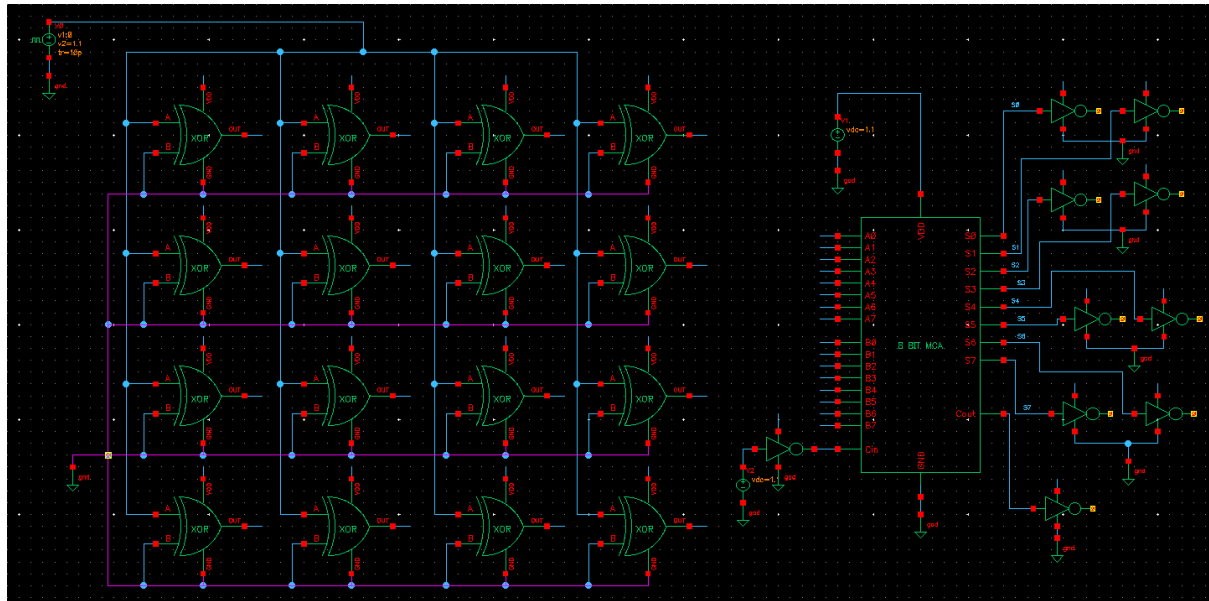


Figure 45 - Optimized Design Max Active Energy Schematic

To maximize switching energy, we simultaneously toggle all A and B inputs from `0` to `1` across both adders. This creates a scenario where every bit—ranging from A0 to A7 and B0 to B7—undergoes a full transition from low to high. By inducing this synchronized state change, we maximize power consumption, capturing the peak switching activity of the system.

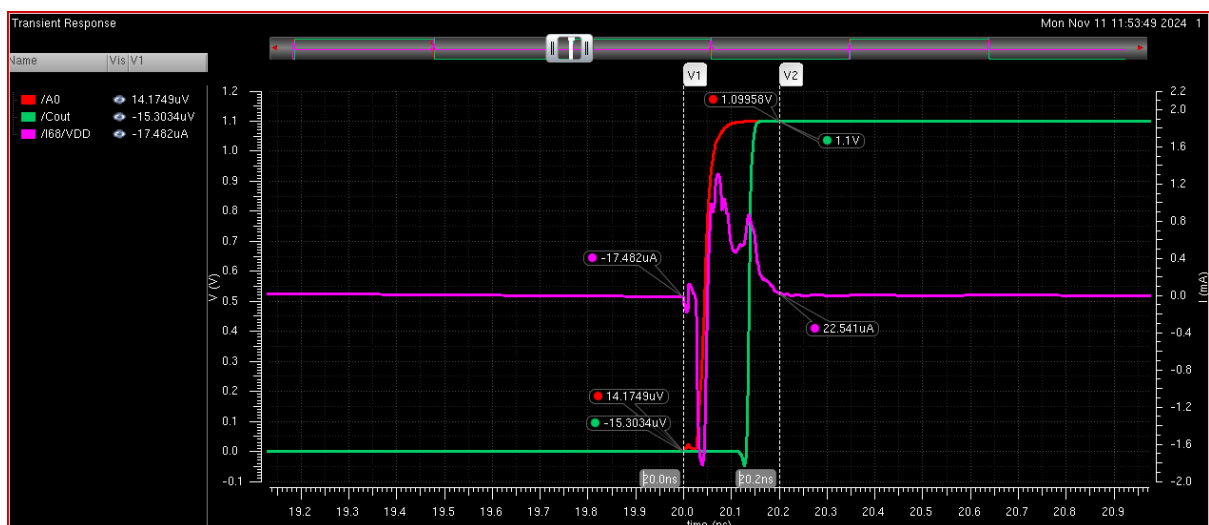


Figure 46 - Optimized Design Max Active Energy Graph

The energy is measured for the waveform where input goes from 0 $\rightarrow$ 1 as depicted in Figure 42.

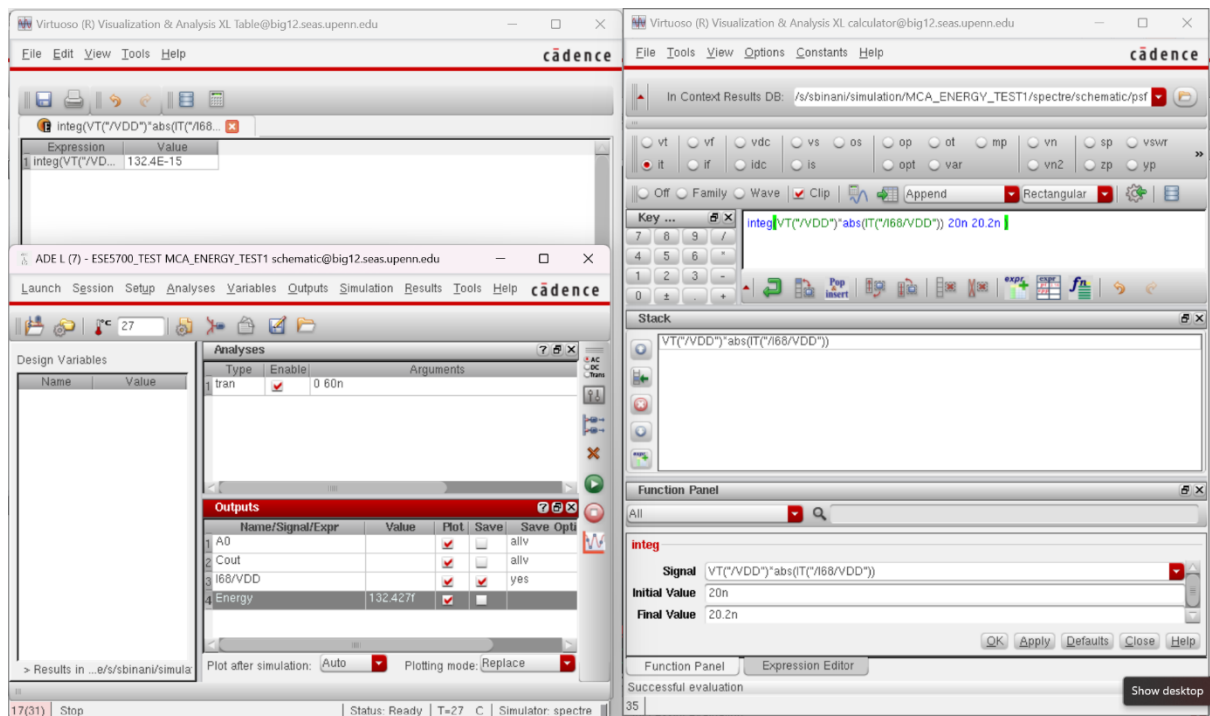


Figure 47 - Optimized Design Max Active Energy

Therefore, the Maximum Switching Energy Value obtained is **132.4 F J**.

2. Average Switching Energy

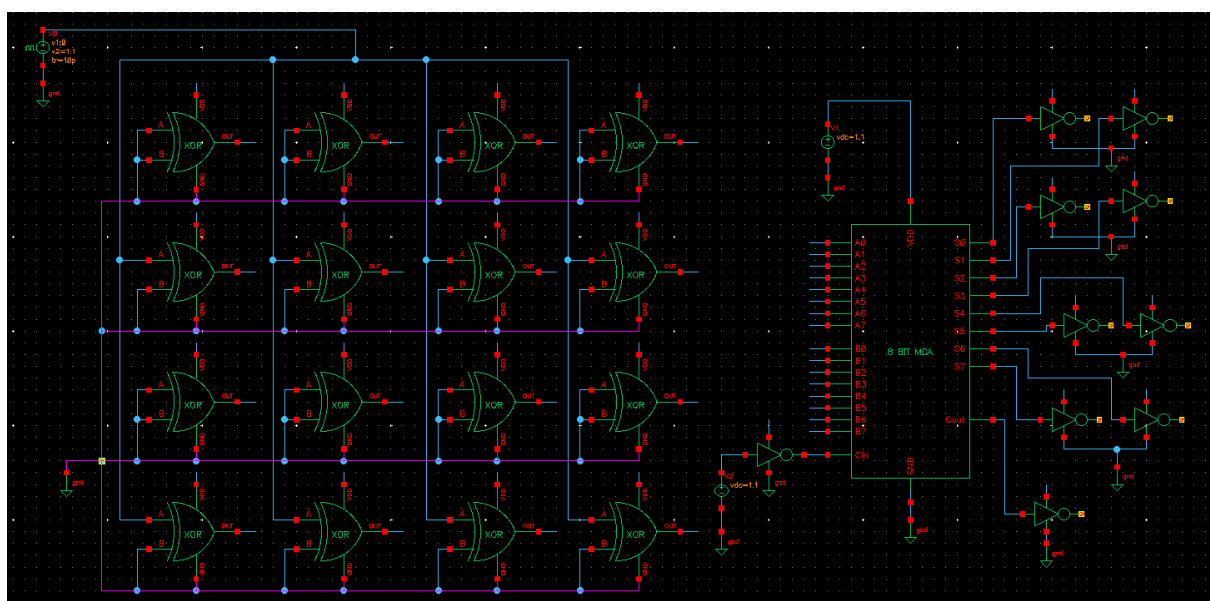


Figure 48 - Optimized Design Average Active Energy Schematic

To calculate the average switching energy, we have configured both A and B inputs as **11110000** for the adders in the first stage. This setup sets the most significant bits (**A4 to A7 and B4 to B7**) to **1** and the lower bits (**A0 to A3 and B0 to B3**) to **0**.

This balanced input pattern ensures partial switching, representing an average case scenario where only half of the bits transition, allowing us to effectively estimate the typical power consumption.

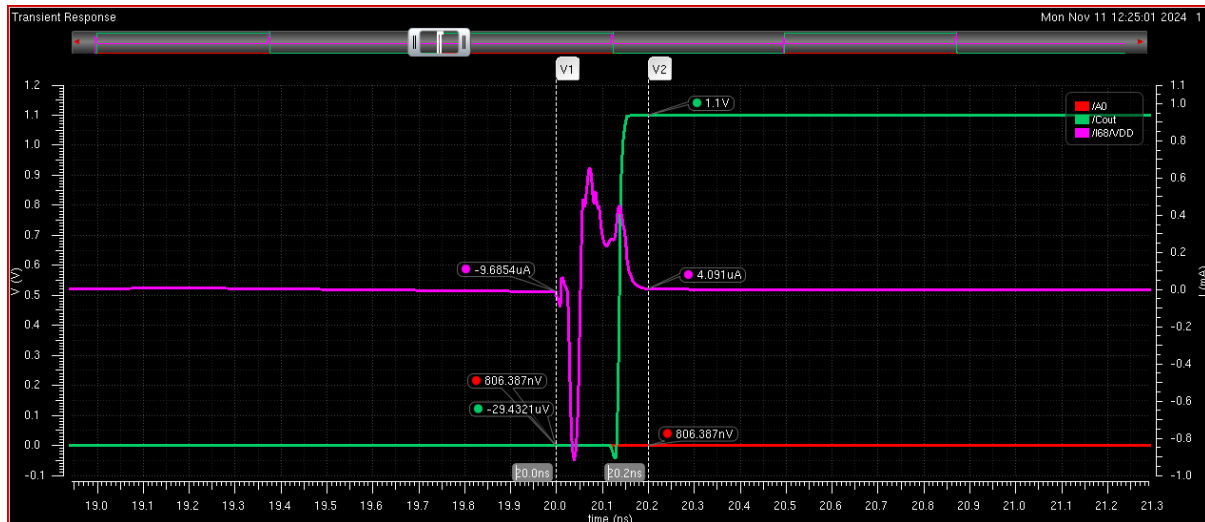


Figure 49 - Optimized Design Average Active Energy Graph

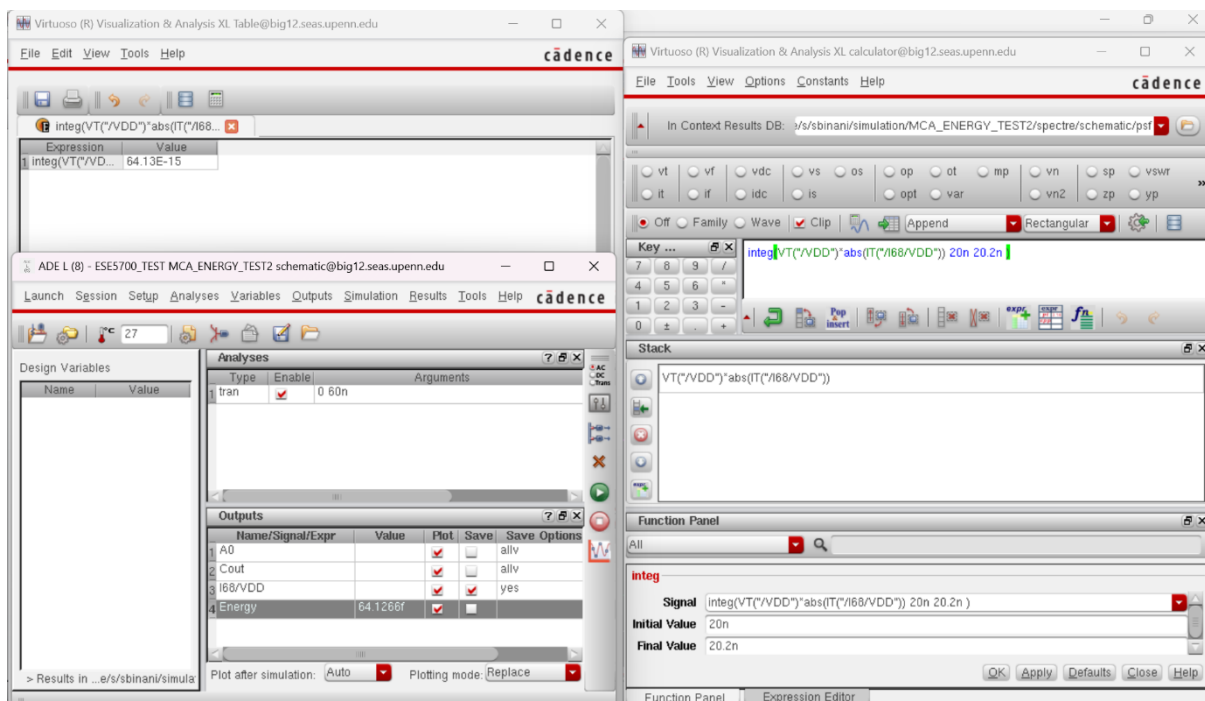


Figure 50 - Optimized Design Average Active Energy

Therefore, the Average Switching Energy Value obtained is **64.13 F J**.

Optimized Leakage Energy Max/Min Cases and Measurements

1. Max Leakage Energy

Max Case – A = 0, B = 0, Cin = 1

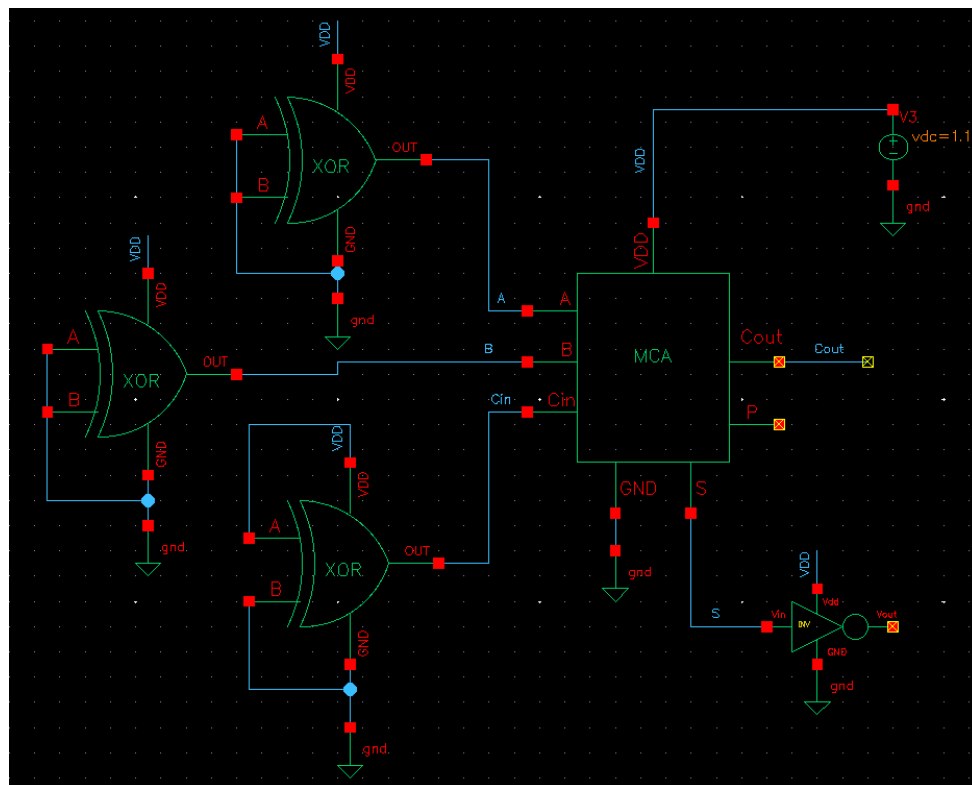


Figure 51 - Test Schematic for Max Leakage Energy (Optimized adder)

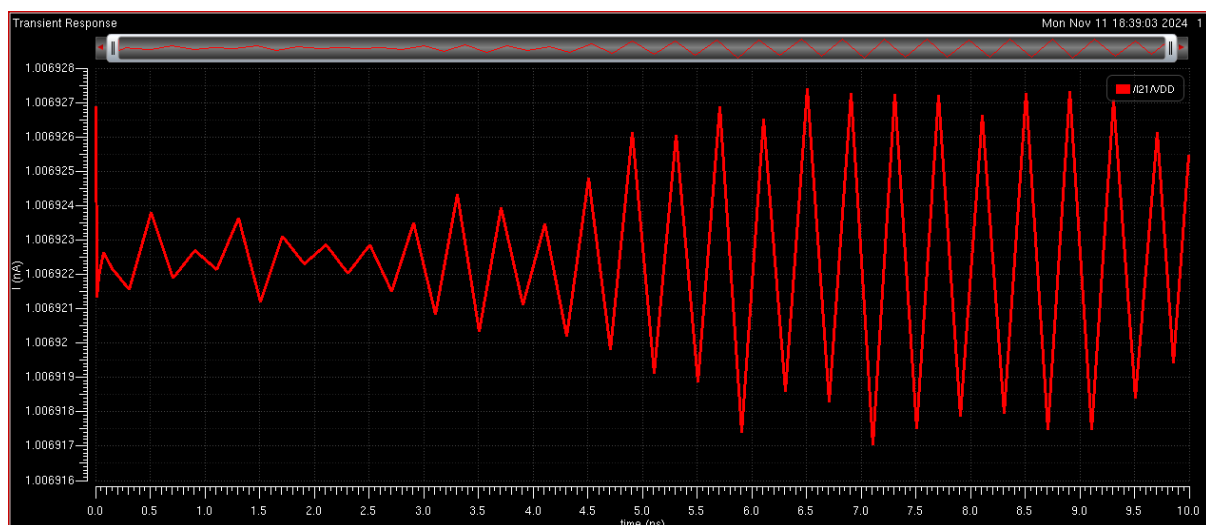


Figure 52 - Graph for Max Leakage Energy 8-bit Optimized Adder

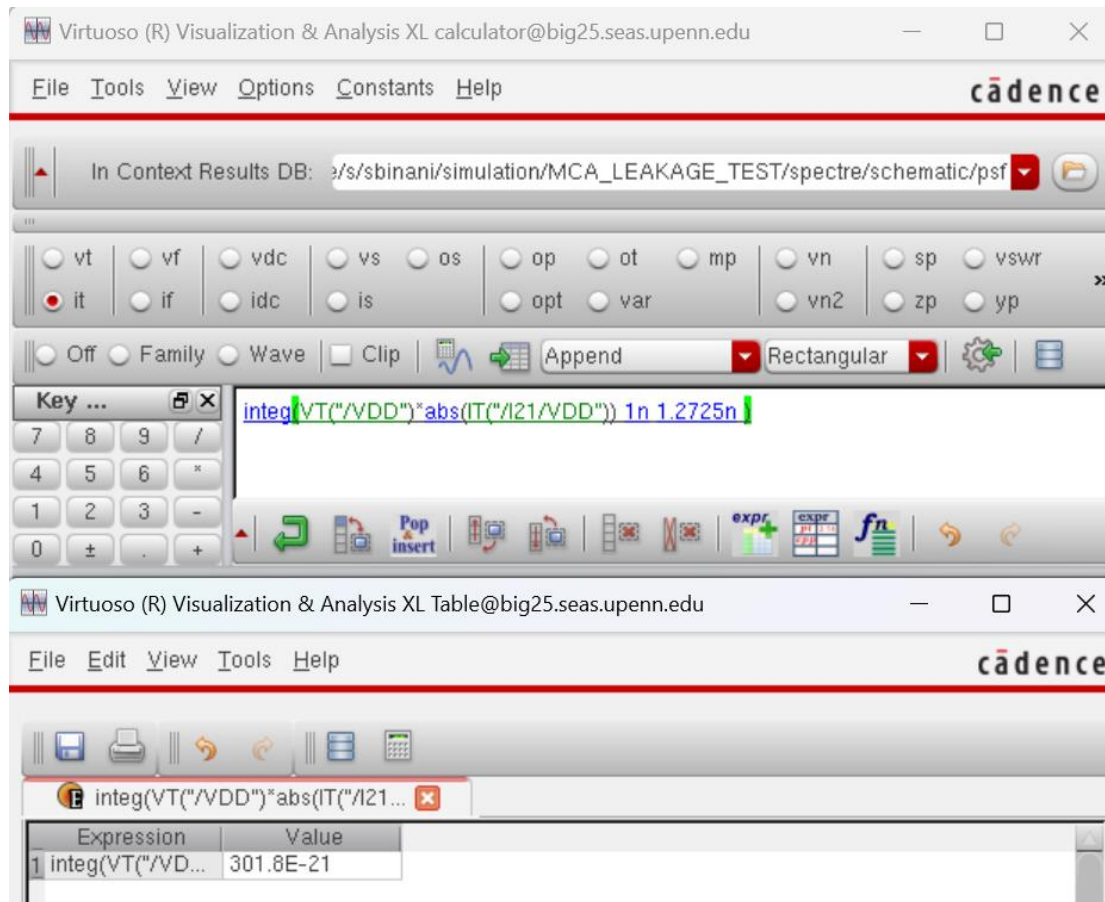


Figure 53 - Max Energy Calculation for Leakage Energy

Therefore, the Maximum Leakage Energy Value obtained for bit-slice is **$301.8 \times 10^{-21} \text{ J}$** .

So, total max leakage energy for 8-bit adder = $301.8 \times 8 = \mathbf{2414.4 \times 10^{-21} \text{ J}}$.

2. Minimum Leakage Energy

Min Case – A = 1, B = 1, C = 1

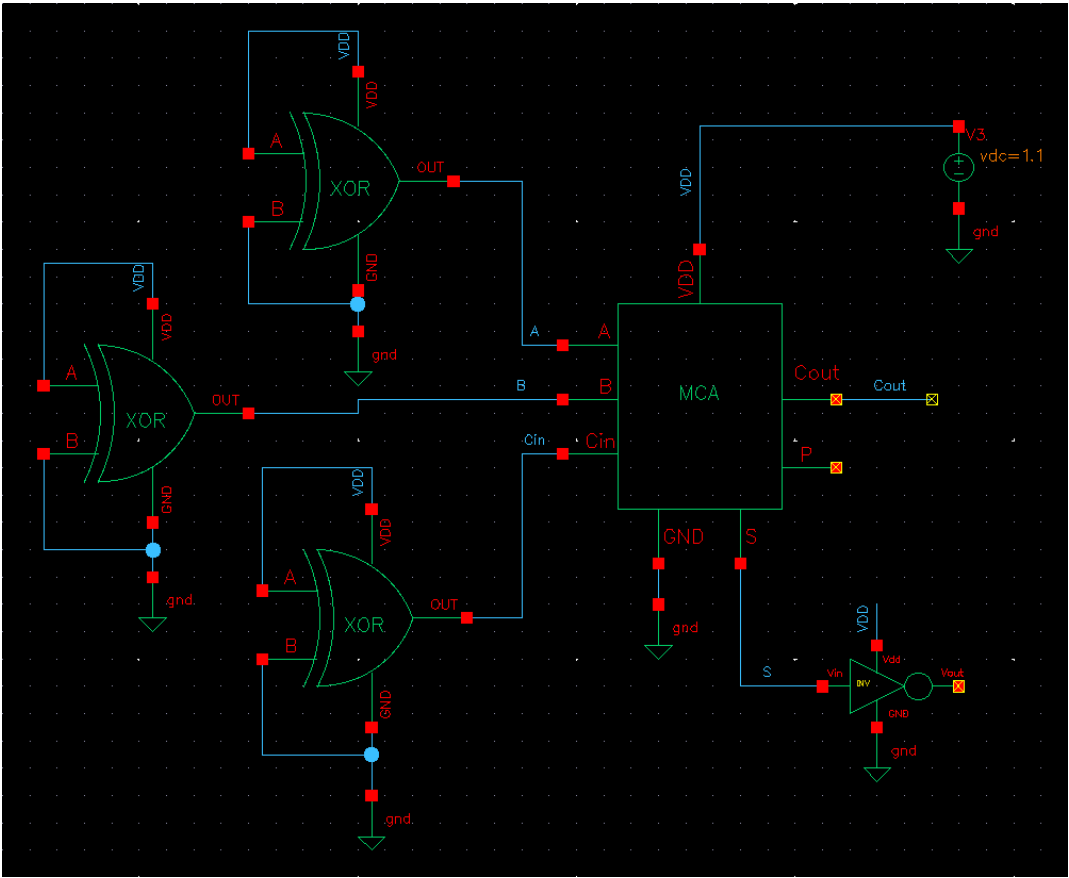


Figure 54 - Test Schematic for Min Leakage Energy (optimized)

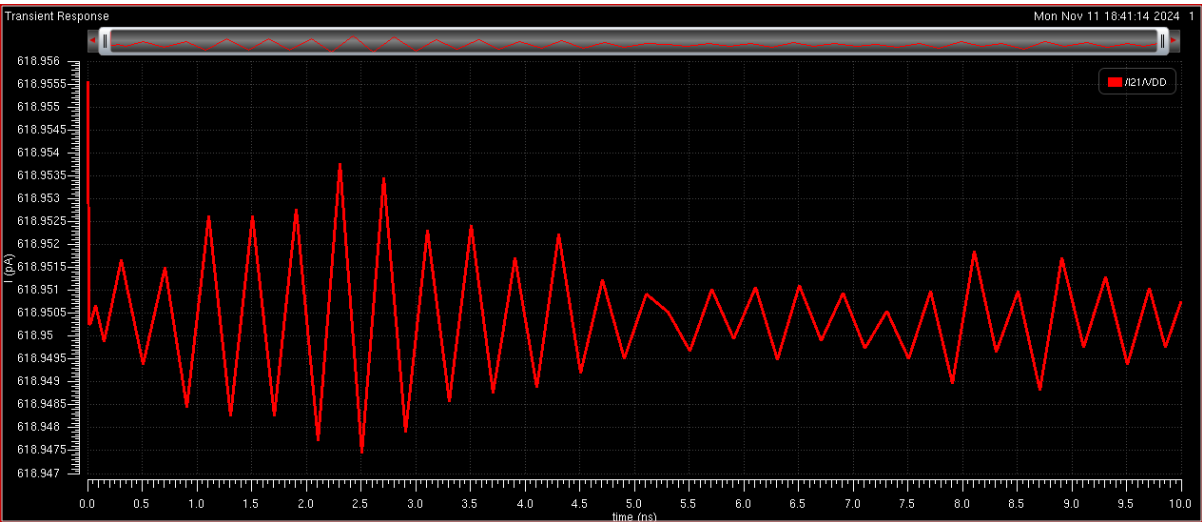


Figure 55 - Graph for Min Leakage Energy

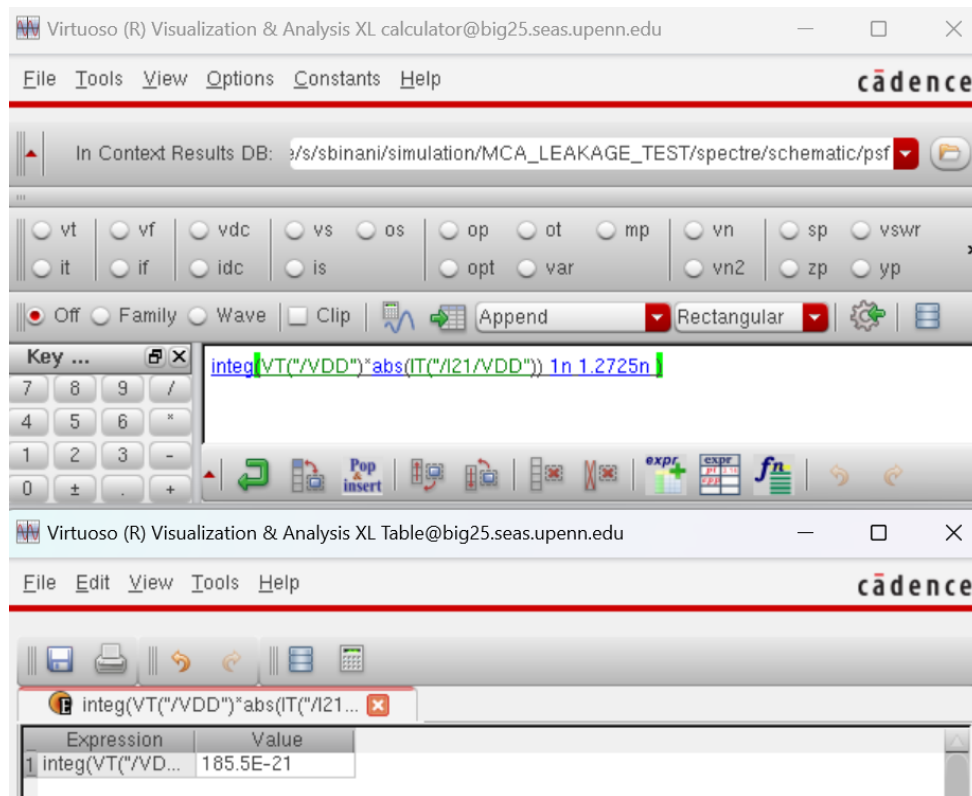


Figure 56 - Min Leakage Energy Calculator Optimized

Therefore, the Minimum Leakage Energy Value obtained for bit-slice is **185.5×10^{-21} J**.

So, total min leakage energy for 8-bit adder = $185.5 \times 8 =$ **1484×10^{-21} J**.

Optimization Process Description

The design process includes transistor-level optimizations to minimize propagation delay and power consumption. We carefully sized transistors and made use of transmission gates, ensuring faster switching while maintaining low power overhead as seen in Table 2.

Table 2 - Optimized Design Sizing PMOS and NMOS

	PMOS	NMOS
XOR2	1.92 u	480 n
AND2	960 n	480 n
AND4	1.92 u	960 n
NOR2	960 n	240 n
INVERTER	480 n	240 n

In our optimized design, we placed two 2:1 multiplexer after every group of four full adders, acting as carry bypass elements for the Manchester Carry Adder. These multiplexers use transmission gates, which can cause a drop in signal voltage. To address this, we added two inverters in series with the multiplexers to restore the signal strength.

In the baseline structure, the worst-case scenario was resolved by implementing carry bypass logic using the Manchester Carry Adder. Specifically, when $p0 = p1 = p2 = p3 = 1$, (where p is the propagate signal), the incoming carry is forwarded immediately to the next block. If this is not the case, the carry is obtained by the propagating through the chain.

This approach enables faster carry propagation and reduces delay for this case.

Design Metric Summary Table with Percentage Improvements

Table 3 - Summary Table between RCA and MCA

Property	Baseline Design	Optimized Design	Percentage Improvements
τ_{pLH}	0.412 ns	0.3371 ns	18.18 %
τ_{pHL}	0.434 ns	0.2545 ns	41.36 %
τ_p (Worst Case Delay)	0.423 ns	0.29575 ns	30 %
Max Switching Energy	23.04 F J	132.4 F J	82.6 %
Average Switching Energy	11.17 F J	64.13 F J	82.6 %
Min Leakage Energy	2414.4×10^{-21} J	1484×10^{-21} J	38.52 %
Max Leakage Energy	1040.8×10^{-21} J	686.8×10^{-21} J	34 %

Based on the design metric summary table (as seen above in Table 1) comparing the 8-bit ripple carry adder (baseline) with a Manchester carry adder (optimized design), significant improvements were achieved in propagation delays.

The optimized design demonstrated a notable **41.36% reduction in high-to-low transition delay** (τ_{pHL}) and an **18.18% improvement in low-to-high transition delay** (τ_{pLH}). The worst-case delay (τ_p) **was reduced by 30%, from 0.423 ns to 0.29575 ns**, indicating better overall performance.

However, this speed optimization came at the cost of increased switching energy consumption, with both **maximum and average switching energy showing an 82.6% increase**.

However, this was offset by a notable reduction in leakage energy; the minimum leakage energy **decreased by 38.52%, and the maximum leakage energy dropped by 34%**. This trade-off reflects a typical scenario in digital design, where enhancing speed often comes at the cost of higher power consumption, but improvements in leakage energy help balance overall efficiency.